Computational Materials Science 261 (2026) 114319

Contents lists available at ScienceDirect

Computational Materials Science

journal homepage: www.elsevier.com/locate/commatsci

Computer vision of ferrite, pearlite, austenite, bainite, and martensite: A deep learning framework toward high-throughput analysis of steel microstructure

Ashutosh Kumar Gupta [a], Subir Gupta [b], Swarup Kumar Ghosh [c], Subhas Ganguly [a,*]

[a] Department of Metallurgical and Materials Engineering, National Institute of Technology, Raipur 492010, India
[b] Department of Computer Science Engineering, Haldia Institute of Technology, Haldia, West Bengal 721657, India
[c] Department of Metallurgy and Materials Engineering, Indian Institute of Engineering Science and Technology, Shibpur, Howrah 711103, India

ARTICLE INFO

Keywords:
Computer vision of microstructure
Steel
Deep learning
Micrograph processing
Phase segmentation

ABSTRACT

Microstructures are 2D images captured in microscopy, which are defined as the structure of a prepared surface of material. Nonetheless, they efficiently result in a structure-property correlation study. However, manual recognition of microscopic data is both human knowledge-intensive and potentially ambiguous, especially in the case of steel. The current study creates a computer vision of important microconstituents of steels such as ferrite, pearlite, austenite, bainite, and martensite, enabling automatic identification of microstructure. Utilizing the Chan-Vese image segmentation method, the current work extracted the patches of phases from the collected 277 microstructures of steels. These phase patches constituted the database for the development of a deep convolutional neural network VGG-19, which depicted 94 % and 86 % accuracy in the training and testing datasets respectively. The article demonstrates the development and prediction framework of *in-silico* microstructure vision. Utilizing a few typical example microstructure data, the article discussed the automatic labelling knowledge intensiveness and ambiguous behavior of the created microstructure vision. The work quantifies the phase fraction from the proposed model and compares it with open source tools using a customized in-house microstructure data set that comprises variation in heat treatments, carbon content, and magnification, which revealed an acceptable level of variation (± 3 %).

1. Introduction

Microstructures are captured by microscopy as a 2D image, which is an extremely approximate measure of the actual structure, it is incredibly successful in explaining the process-structure-property relationship of materials. Because of this, it has been perpetually used in the design and development of metallic alloy systems for a long time [1–3]. Nevertheless, manual recognition of microscopic image data is not only a challenging and tedious task but also a knowledge-intensive and potentially ambiguous process. It is furthermore difficult in the case of steel microstructure, which encompasses a diverse variant of microconstituent with a variety of complex morphologies in each variant [4,5]. Therefore, it is a potential technology barrier in the automation and digitization of steel processing industries, and thus, the implementation of Industry 4.0. Development of a metallographically knowledge-intensive computer vision technology can be a breakthrough to counteract the issue [6,7]. The microstructure analysis objectives mainly include the recognition of individual phases, measurement of phase fraction, average grain size, shape, phase distribution, and grain boundary in the case of already known alloys' microstructure [8,9]. The qualitative and quantitative measurements of these morphological features are approximated through mathematical models to map various material properties. However, microstructures of newly explored alloys need extensive experimentation and characterization [10–12].

In digital microstructure analysis, precise image segmentation is a key method to extract phase information and is inevitable for the digital analysis of microstructures to achieve automatic analysis [13–15]. Conventional segmentation techniques like watershed and Otsu thresholding provide several possibilities for automation. Nevertheless, these techniques work well on microstructures that have easily identifiable phases in optical microstructure images, for example, dual-phase steels, which predominantly consist of distinct phases like ferrite and

* Corresponding author.
E-mail address: sganguly.mme@nitrr.ac.in (S. Ganguly).

https://doi.org/10.1016/j.commatsci.2025.114319
Received 27 June 2025; Received in revised form 22 September 2025; Accepted 13 October 2025
Available online 21 October 2025
0927-0256/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

martensite [16,17]. Decost and Holm [18] introduces a computer vision approach for the automated analysis and classification of microstructural image data, leveraging the support vector machine and bag of visual features (BoVF). The study acknowledged limitations such as the lack of incorporating physical scale information and spatial relationships into the image representation. These techniques become ineffective where it is difficult to discriminate between phases, for example, when investigating complex microstructures under secondary electron microscopy (SEM). The revolution of machine learning (ML) has emerged as a vital tool in material microscopic image analysis since the development of deep learning technology. Deep learning (DL) has become a standard technique in computer image processing by utilizing neural network models to automatically learn strong feature representations from data through optimizing the loss function [19–21].

The ability of deep learning approaches to precisely define object regions and extract complex object boundaries has completely transformed the field of image segmentation. Deep learning techniques for image recognition, detection, and segmentation are frequently employed in materials science to enhance qualitative results and support microscopic image analysis [22–24]. In the context of image analysis, Convolutional Neural Networks (CNN) are at the forefront of this revolution. End-to-end segmentation of material microscopic images has been made possible by dense Fully Convolutional Networks (FCNs) by adopting CNN for pixel-wise predictions [25,26]. FCNs sequentially connect and combine convolutional, pooling, and sampling layers to acquire hierarchical information at various scales and produce comprehensive segmentations. Many sophisticated structures, like DeepLab [27], SegNet [28], and PSPNet [29] have been proposed based on the FCNs. By using advancements in convolutions, dilated convolutions, and pyramid pooling, these models improve their capacity to handle different object scales and capture context. The next step saw the creation of U-Net architectures [30], which became widely used in material and medical image processing by introducing skip connections to integrate high-level context with fine-grained spatial information. Chaurasia et al. [31] introduces an innovative training methodology for segmenting ferrite-pearlite microstructures in steel using a convolutional neural network (CNN) U-net architecture. The primary challenge addressed is the common requirement in CNN-driven image segmentation, which are typically unavailable. It creates the foundation for more complex designs and establishes the baseline for image segmentation [32,33].

The advanced high-strength steels, a class of the most industrially important steel of recent times, are mainly designed with a microstructure that comprises ferrite, pearlite, bainite, martensite, or retained austenite [34,35]. Such complex phase steels contain two or more of these micro-constituents. The two-phase steels are very common; however, multiphase steels have also existed [36,37]. Although the SEM microstructure of such steels exhibit heterogeneity in gray scale value in their images. But using conventional gray scale segmentation, the scope of differentiating all these features in such microstructures is very limited. Methods for digital analysis of steel microstructure have been developed in association with the advancement of computer vision using machine learning, especially deep learning. DeCost et al. [38] introduces a novel approach to quantitative microstructure analysis using deep convolutional neural networks (CNNs), specifically focusing on ultra-high carbon steel (UHCS). The research aims to overcome the limitations of traditional manual or semi-automated metallography, which are often subjective, labor-intensive, and limited by low-level image features. Ali Riza et al. [39] proposed a deep learning approach using U-Net architecture for segmentation of bainite in steel microstructure. Xiu Liu et al. [40] recently demonstrated a procedure with transfer learning using convolutional neural networks (CNNs) in the segmentation of carbon steel microstructures. Tsutsui et al. [41] adopted gray-level co-occurrence matrix (GLCM) for texture analysis of the SEM images, same as Gupta et al. [42] successfully adopted a multi-layer perceptron (MLP) model that includes a gray level co-occurrence matrix for feature extraction and was used to recognize distinct phases in steel microstructure. Azimi et al. [43] used a deep learning model for the classification of low-carbon steel microstructure. In this work, a novel method called pixel-wise segmentation using FCNN supported by a max-voting scheme was proposed, enabling the achievement of 93.94 % accuracy. Bachmann et al. [44] developed a machine learning model to automatically and reproducibly classify the complex microstructures of quenched and tempered steel. In this work they implemented a patch-wise classification scheme to quantify entire micrographs and achieve a *de-facto* segmentation of microstructures. The model claimed 88.8 % and 93.7 % prediction accuracy in light microscopy and scanning electron microscopy images respectively, overcoming the traditional subjective evaluation methods. Muller et al. [45,46] used machine learning technique for automated and reproducible classification of the multiphase steels. The textural and morphological parameters were considered for feature extraction and classified using support vector machine. DeCost et al. [47] establishes a dataset for complex, hierarchical, and technologically relevant microstructures, demonstrating that appropriately pooled and encoded local features (SIFT) and domain-transferred deep CNN representations can achieve high classification accuracy, exceeding 95 %. The presented method was effective for classification, but not able to fully capture the nuances required for precise quantitative analysis. The deep learning research on plausible segmentation and quantification of steel microstructures is ongoing.

In this work, we have developed a methodology of patch-wise semantic segmentation by the combined use of the Chan-Vese algorithm along with the VGG-19 deep network to create the digital vision of ferrite, pearlite, austenite, bainite, and martensite. It produces a fast and accurate two-phase steel microstructure image analysis framework. To deal with the large resolution and complex morphological parameters of the microstructure, the image was split into many patches according to the phase present in the micrograph. The microstructure segmentation process was performed using Chan-Vese segmentation, and the patches were created from the largest bounding rectangle of phases present in the steel microstructure. The present article contributes to the implementation of Chan Vese segmentation for creating phase patches instead of just using a regular grid for patch extraction as proposed earlier [44]. Moreover, the study used in-house generated own experimental data set along with literature data enhancing the model validation in a more realistic environment. The model prediction results have been discussed with a set of in-house generated SEM micrographs of a few steels in the article. The schematic description of the overall framework for high-throughput analysis of the SEM microstructure of steel is presented in Fig. 1.

2. Materials and methods

2.1. SEM microstructure dataset

A steel microstructure database was created by an extensive review of published literature. Relevant data were meticulously gathered from various scientific articles, research papers, and industrial reports that detailed the microstructural characteristics and phase compositions of steels belonging to the advanced high-strength steels. The ground reality for labeling micrographs was done as they are reported in the literature. The literature sources were carefully selected based on their relevance, rigor, and the quality of their experimental methodologies. Data collection included the volumetric fractions of different phases, chemical compositions, and heat treatment processes. The size of the primary data set was approximately 450 SEM images; however, after preprocessing and cleaning, 89 micrographs were used in the present work (sources are listed in appendix). Also, a benchmark SEM (model: Zeiss EVO 18, Germany) micrograph dataset was prepared through in-house experimentations, comprising 188 micrographs. All metallographic images were annotated and marked by two experienced experts to ensure the consistency of the dataset. Few Transmission Electron

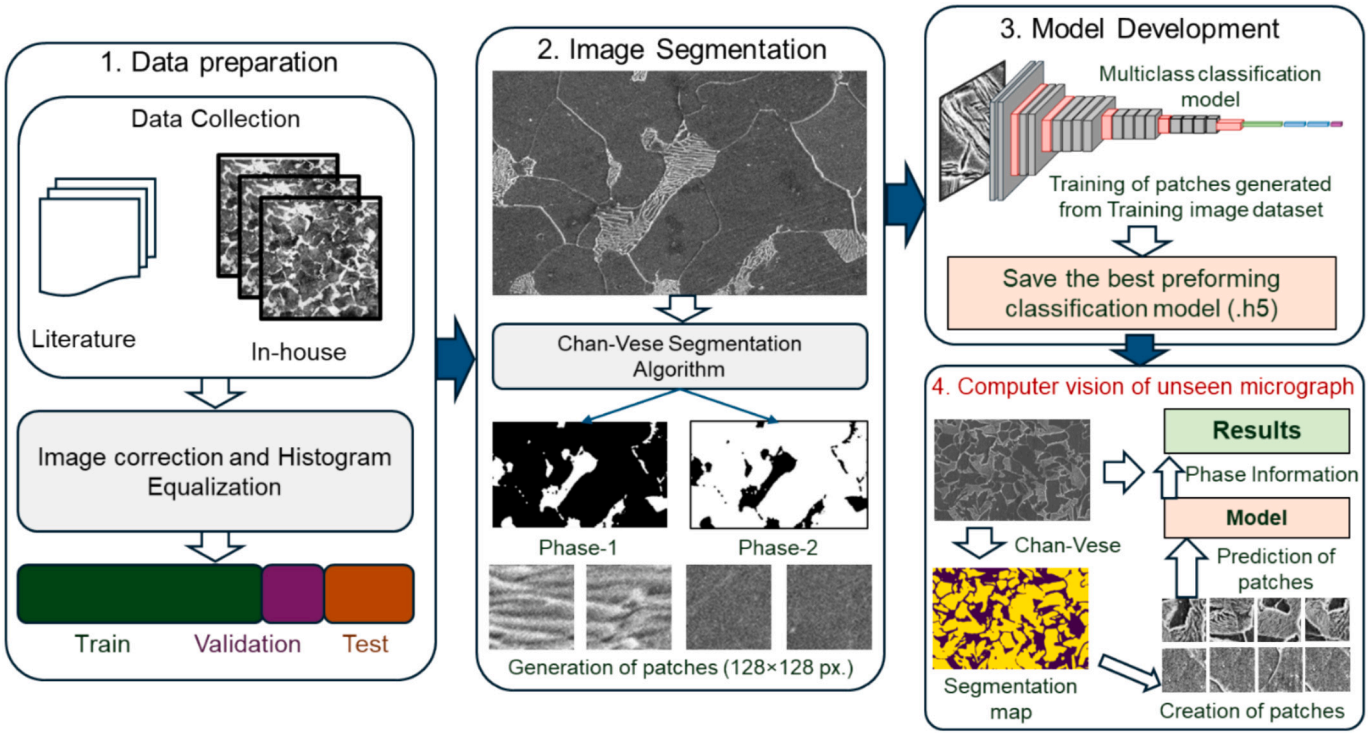


Fig. 1. Schematic representation of image segmentation and classification model framework proposed for digital recognition of phase constituents such as ferrite, pearlite, austenite, bainite, and martensite in SEM micrograph image of steel. The first block demonstrates the data collection splitting while the second block explains the segmentation of microstructure and generation of patches of segmented phases. The latter block represents the DL model training with phase patches and prediction of phase constitution in an unseen microstructure.

Microscope (TEM) studies were also done to ensure the annotation of in-house SEM micrographs used as the case studies for prediction analysis of the proposed high-throughput microstructure analysis model framework.

2.2. Data preprocessing and partitioning strategy

Data preprocessing is a critical step in ensuring the quality and reliability of the image dataset used for deep learning analysis. In this study, preprocessing involved histogram equalization and the elimination of suboptimal images, such as optical microscopic images and blurred images. Histogram equalization was performed to standardize the basic intensity characteristics across all images, thereby minimizing the influence of variable illumination on the segmentation and classification results. Let us consider the variable I denotes the intensities of a grayscale micrograph to be processed. For an image, I should be in range $[0,L-1]$ with $I = 0$ representing black and $I =$ L-1 (i.e., 255) representing white [48]. If the intensity distribution of an image is taken as a histogram $h(I)$, then the cumulative distribution function for each gray value (x) can be obtained as:

$$cdf(x) = \sum_{I=1}^{x} h(I) \tag{1}$$

Then, the general histogram equalization $h(v)$ (is)

$$h(v) = round\left(\frac{cdf(x) - cdf_{min}}{(M \times N) - 1} \times (L-1)\right) \tag{2}$$

where cdf_{min} the minimum non-zero value of the cumulative distribution function. M, N are the length, width of the gray-scale image. L is the number of gray levels used (i.e., 256) [48].

Additionally, images with significant blur or those captured through optical microscopy, which did not meet the resolution and clarity requirements, were systematically excluded from the dataset. The SEM dataset was divided into three parts namely training (70 %), testing (20 %), and validation (10 %) to avoid data leakage. All splits were performed at the SEM-image level, patches were derived from 277 unique micrographs, and every patch is labeled with its source image ID. Training, validation and test sets were created by partitioning these image IDs, not patches so that no patches from the same SEM micrograph appear in more than one set. The testing dataset was stored separately only for the evaluation of the model. Data augmentation was applied after splitting and released the split indices to ensure reproducibility. This rigorous preprocessing protocol ensures that the remaining images are of high quality and suitable for subsequent analytical procedures, enhancing the robustness and accuracy achieved through the findings of this study.

2.3. Microstructure segmentation and patch creation

Semantic segmentation is a powerful tool of deep learning to segment complex image problems that can't be resolved using traditional techniques (i.e., thresholding). It may be possible by a more recent technique, pixel wise classification of images which rely on each pixel of the image is labeled to a class, and thereby, the whole image is segmented. Moreover, pixel-wise classification has capability to segment images with more than two classes in the same segmentation step. U-net, SegNet and different versions of DeepLab are some examples of pixel wise classification approach. To train such semantic segmentation models, ground truth (i.e., masks) that explicitly allocate each pixel to a particular class must be prepared. In contrast to pixelwise segmentation we have introduced a patch-creation method that integrates the Chan-Vese segmentation model with the largest area contour model for fast and accurate phase detection. This approach enables the automatic creation of patches corresponding to distinct phases. Initially, the Chan-Vese image segmentation technique was employed to identify distinct clusters within the image. The Chan-Vese method is based on the minimization of energy [49]. Let us consider dual-phase steel

microstructure, which is formed of two regions – martensite and ferrite, having intensity values of I_i and I_o for inside and outside of the martensite region respectively. The boundary of the actual martensite region is denoted by C_0. Inside the martensite region C_0, the mean intensity is $V_0 \approx I_i$, and outside the C_0, the mean intensity is $V_0 \approx I_0$. The eq. (3) represents the fitting term.

$$F_1(C) + F_2(C) = \int_{inside(C)} |V_0(x,y) - k_1|^2 dxdy + \int_{outside(C)} |V_0(x,y) - k_2|^2 dxdy \tag{3}$$

where F_1 and F_2 are fitting terms. $V_0(x, y)$ is the intensity value of a micrograph pixel. C is a sought contour that separates martensite and ferrite. k_1 and k_2, which depend on C, are average intensity value constants inside and outside the curve respectively. Let us consider a level set function $\phi(x, y)$ which describes the different areas of the segmented micrograph (say $\Re$), then k_1 and k_2 will be:

$$\begin{cases} k_1 = average V_0(x,y) \; in \; \{\phi \geq 0\} \\ k_2 = average V_0(x,y) \; in \; \{\phi < 0\} \end{cases} \tag{4}$$

The boundary of the martensite is the minimizer of the fitting term. Infimum (greatest lower bound) is shown as inf in the following eq. (5)

$$\inf_C \{F_1(C) + F_2(C)\} \approx 0 \approx F_1(C_0) + F_2(C_0) \tag{5}$$

If the curve C is inside the martensite, then $F_1(C) \approx 0$ and $F_2(C) > 0$. If the curve C is outside the martensite, then $F_2(C) \approx 0$ and $F_1(C) > 0$. If the curve is both inside as well as outside the martensite, then $F_1(C) > 0$ and $F_2(C) > 0$. When the curve C is on the boundary of the martensite, the fitting energy is minimized (i.e., $C = C_0$). In this case, both the fitting energy terms $F_1(C)$ and $F_2(C) \approx 0$.

In the Chan-Vese model, the above fitting term is minimized, and the length of the curve (C) and the area of the region inside C are added as regularization terms. *F (k_1, k_2, and C)* is introduced as an energy functional, which is defined as

$$\begin{aligned} F(k_1,k_2,C) = {} & \mu \bullet Length(C) + \eta \bullet area(inside(C)) \\ & + \lambda_1 \int_{inside(C)} |V_0(x,y) - k_1|^2 dxdy \\ & + \lambda_2 \int_{outside(C)} |V_0(x,y) - k_2|^2 dxdy \end{aligned} \tag{6}$$

where $\mu \geq 0$, $\eta \geq 0$, and $\lambda_1, \lambda_2 > 0$. The values of this parameter are fixed at $\lambda_1 = \lambda_2 = 1$ (to give equal importance to both inside and outside of the contour C), $\eta = 0$ (to make it unbiased to any area-size of the contour C), as proposed by Chan and Vese [49]. Eq. (6) can be minimized by taking a level set function ϕ (x, y) which describes the different areas of the segmented micrograph (say $\Re$). The level set function (i.e., $\phi(x, y)$) takes a value greater than 0 for inside the martensite contour, less than 0 for outside the martensite, and equal to zero on boundaries. H is the Heaviside function, and δ is its derivative. The regularization of H is used and defined as follows:

$$H(z) = \begin{cases} 1 \; if \; z \geq 0 \\ 0 \; if \; z < 0 \end{cases} \tag{7}$$

Combining eqs. (6) and (7)

$$\begin{aligned} F(k_1,k_2,\phi) = {} & \mu \bullet \int_{\Re} \delta(\phi(x,y)) \bullet |\nabla\phi(x,y)| dxdy + \eta \bullet \int_{\Re} H(\phi(x,y)) dxdy \\ & + \lambda_1 \int_{\Re} |V_0(x,y) - k_1|^2 H(\phi(x,y)) dxy \\ & + \lambda_2 \int_{\Re} |V_0(x,y) - k_2|^2 (1 - H(\phi(x,y))) dxy \end{aligned} \tag{8}$$

Keeping ϕ fixed and minimizing the value of $F(k_1, k_2, \phi)$ concerning the constants k_1, k_2. The expression for k_1, and k_2 are as follows:

$$\begin{aligned} k_1(\phi) &= \frac{\int_{\Re} V_0(x,y) \bullet H((\phi(x,y)) dxdy}{\int_{\Re} H(\phi(x,y)) dxdy} \; k_2(\phi) \\ &= \frac{\int_{\Re} V_0(x,y) \bullet (1 - H(\phi(x,y))) dxdy}{\int_{\Re} (1 - H(\phi(x,y))) dxdy} \end{aligned} \tag{9}$$

The values of $k_1(\phi)$ and $k_2(\phi)$ have the meaning of average brightness of the micrograph over the areas that are regarded as martensite ($\phi \geq 0$) and ferrite ($\phi < 0$), respectively, at the time of image segmentation. By providing k_1 and k_2 with a fixed value and minimizing $F(k_1, k_2, \phi)$ with respect to ϕ, the associated Euler-Lagrange equation may be obtained as follows,

$$\begin{aligned} \frac{\partial\phi(x,y)}{\partial t} &= \delta(\phi)\left[\mu \; div\left(\frac{\nabla\phi}{|\nabla\phi|}\right) - \eta - \lambda_1(V_0 - k_1)^2 + \lambda_2(V_0 - k_2)^2\right] \\ &= 0 \; in \; \Re \end{aligned} \tag{10}$$

Eq. (10) is used for an iterative calculation of $\phi(t, x, y)$. Thus, a termination criterion (50 in our case) is required. Each iteration step *n* is followed by a calculation of measure Q for the stationarity of the segmentation:

$$Q = \sum_{|\phi_n| < 0.5} |\phi_n \phi_{n+1}| \tag{11}$$

The iteration of micrograph segmentation is terminated when Q falls below a predefined threshold value. The condition for summation $|\phi_n| < 0.5$ ensures that only values near the contour C are considered. To extend the evolution of $\phi(x, y)$ to the entire micrograph $|\nabla\phi|$ can be used instead of $\delta(\phi)$ in eq. (10). In this work of segmentation, wider results were obtained using $|\nabla\phi|$. Since the initialization $\phi_0(x, y)$ has a significant influence on the segmentation results, it must be chosen carefully for ϕ (0, x, y). The image curvature term $div\left(\frac{\nabla\phi}{|\nabla\phi|}\right)$ of Eq. (10) can be represented by the following eq. (12)

$$div\left(\frac{\nabla\phi}{|\nabla\phi|}\right) = \frac{\phi_{xx}\phi_y^2 - 2\phi_x\phi_y\phi_{xy} + \phi_{yy}\phi_x^2}{\left(\phi_x^2 + \phi_y^2\right)^{\frac{3}{2}}} \tag{12}$$

The number of iterations was fixed to 50 for all the cases irrespective of their type. The detailed step-by-step procedure has been depicted in Fig. 2. Subsequently, the largest area bounding box is utilized to crop the image to a specific region associated with a particular phase. This method effectively delineates and isolates different phases within the image by combining advanced segmentation and contour detection methodologies. The availability of hardware like GPU and CPU plays a crucial role in classification and segmentation problems based on the resolution of the microstructure image. Due to the large requirement of GPU memory and time consumption, it is difficult to fully fit high-resolution microstructure images. Therefore, the microstructure image has been cropped (multiple in number) to a smaller size at the point of interest of the microstructures. A previous study by Backmann et al. [44] suggests that patch size of 128 × 128 pixel contains enough information to be able to assign a well-founded and objective label for the ground truth. The bounding box method was used to crop the regions from the image, and then three patches were created with a reasonable size of 128 × 128 pixels by default from a random location where the probability of that phase was high. By doing this, a total of around 945 patches were created: 168 Austenite, 129 Bainite, 204 Ferrite, 279 Martensite, and 165 Pearlite. Automated segmentation outputs were manually inspected by the domain expert. In cases where the predicted boundaries deviated from ground truth, corrections were applied using the rerun the patch creation tool.

To mitigate class imbalance and improve model generalization, data augmentation was applied to the extracted more image patches. The

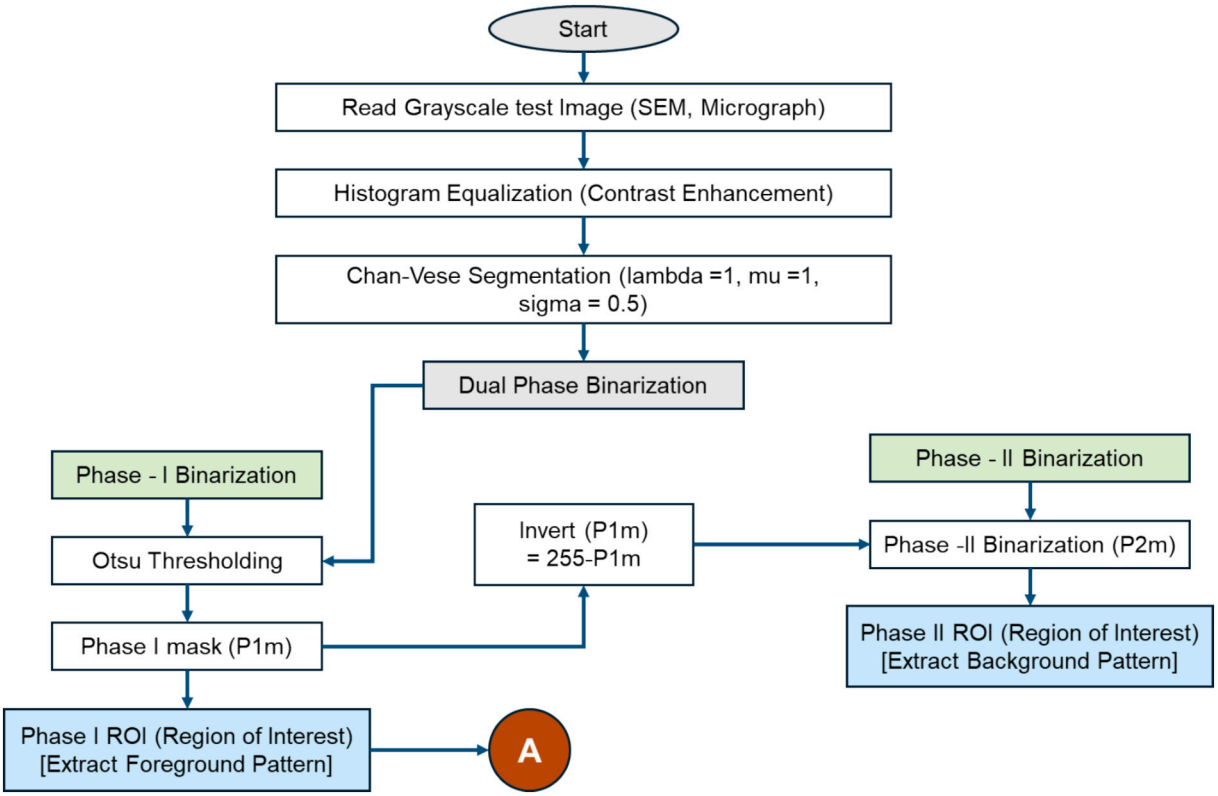


Fig. 2. The schematic procedure to segment the microstructure using Chan-Vese segmentation technique after that extraction of region of interest.

augmentation pipeline included geometric transformations such as horizontal and vertical flipping, random rotations within ±20°, shearing with a shear range of 0.1, and zooming within a range of 0.2. These operations were applied stochastically during training to increase the effective size and variability of the dataset while preserving the essential microstructural features. The example patches of each phase, counts of the original patch along with the total patches augmented are collectively shown in Table 1.

2.4. Label encoding

In this study, we address a multiclass classification problem involving five distinct phases: ferrite, pearlite, austenite, bainite, and martensite. Unlike binary classification, which can be straightforwardly represented using a binary variable with two states (zero and one), multiclass classification necessitates a more sophisticated mathematical representation of the classes. This may be possible using the label binarizer method, specifically one-hot encoding, which generates a unique binary array for each classification class. The encoding of the classification phases was performed manually to label the classes accurately, and the resultant one-hot encoding for the phases is summarized in Table 1.

2.5. Phase classification models

The convolutional neural networks are predominantly used in different image classification case studies due to their ease of implementation and learning abilities at various layers. The CNN models need to be efficient and quick for the detection of the phases in steel analysis. Therefore, the three commonly used CNN-based architectures, like VGG-19, Resnet-50, and EfficientnetB3, are selected to train augmented patches for a comparative analysis of the models. However, some modifications have been made to these original networks to adapt to our work. Two additional fully connected layers were added to extract features with high resolution, and one dropout layer was added in the network to avoid overfitting. The last layer of the classification model was SoftMax, and the binary cross-entropy was selected as the loss function in this study. Adam was used as optimizer for the loss function, and for the model hyperparameter tuning grid search strategy was adopted. The search space included: learning rate (0.01 to 0.0001), batch size (5 and 10), and dropout probability (0.1–0.6). The found optimum condition has been discussed in section 3.1 Phase classification model development.

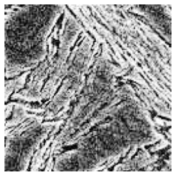
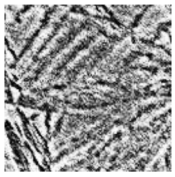
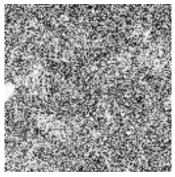
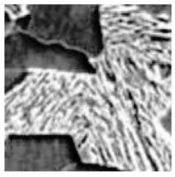
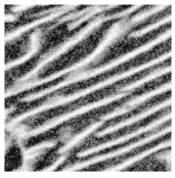
2.5.1. VGG – 19

VGG-19 is a convolutional neural network (CNN) based architecture designed for image classification, and it was proposed by the Visual Geometry Group (VGG) from the University of Oxford. VGG-19 is primarily known for its depth and simplicity, making it a popular choice for various image classification and recognition tasks. It consists of a total of 19 layers with learnable weights in which 16 layers are convolutional layers, and 3 layers are fully connected layers followed by 5 max-pooling layers and the last layer is a SoftMax layer as shown in Fig. 3. The main characteristic of VGG-19 is the use of very small convolutional filters of size 3 × 3 with a stride of 1 and padding of 1.

2.5.2. Resnet50

ResNet-50 [50] is a variant of residual neural network (ResNet) architecture which consists of 50 deep convolutional neural networks to overcome the vanishing gradient problem. In the ResNet model, residual elements with skip connection are primarily incorporated into deep architecture as illustrated in Fig. 4 (a). Its core building block is the bottleneck residual block, consisting of three convolutional layers: a 1 × 1 convolution for dimensionality reduction, a 3 × 3 convolution for feature extraction, and another 1 × 1 convolution for dimensionality restoration as shown in Fig. 4(b). Each block incorporates a skip connection that adds the input directly to the output, enabling stable gradient flow during backpropagation and allowing networks to scale

Table 1
A detailed description of the phase patches dataset and their one hot encoding representation of phases used in deep neural network training.

Image (patch)	Created	Total patch after Augmentation	Phase	Label
	168	619	Austenite	[1,0,0,0,0,]
	129	608	Bainite	[0,1,0,0,0,]
	204	613	Ferrite	[0,0,1,0,0,]
	279	608	Martensite	[0,0,0,1,0,]
	165	603	Pearlite	[0,0,0,0,1,]

much deeper than before. The architecture begins with convolution and pooling layers, followed by multiple stacked residual blocks, and concludes with global average pooling and a fully connected classification layer. The network learns and chooses how the gradient propagates on its own, therefore even if network depth increases, the error will not increase. The ResNet method primarily learns the residual relationship between input x and output $y : F(x) = y - x$. It finally takes $F(x) + x$ as the output target. The ResNet-50 architecture is splitted into five substages. Stage 0 in ResNet-50 structure is rather straightforward and may be generally treated as input preprocessing stage. The bottleneck structures that make up the final four phases are comparatively identical. Three bottleneck structures are present in stage 1, and four, six and three bottleneck are present in the subsequent three phases, respectively.

2.5.3. *EfficientNetB3*

The Efficientnetb3 [51] is variant of the Efficientnet series, is a convolutional neural network that achieves high accuracy while being computationally efficient through a compound scaling method (Fig. 5). Unlike traditional scaling approaches that only increase network depth or width, EfficientNet uniformly scales depth, width, and input resolution using a fixed compound coefficient. Its architecture is built upon MBConv (Mobile Inverted Bottleneck Convolution) blocks (Fig. 5(b and c)), which integrate depthwise separable convolutions, expansion and projection layers, squeeze-and-excitation (SE) modules, and inverted residual connections. EfficientNet-B3 represents a scaled-up version of the baseline EfficientNet-B0, increasing both resolution and network

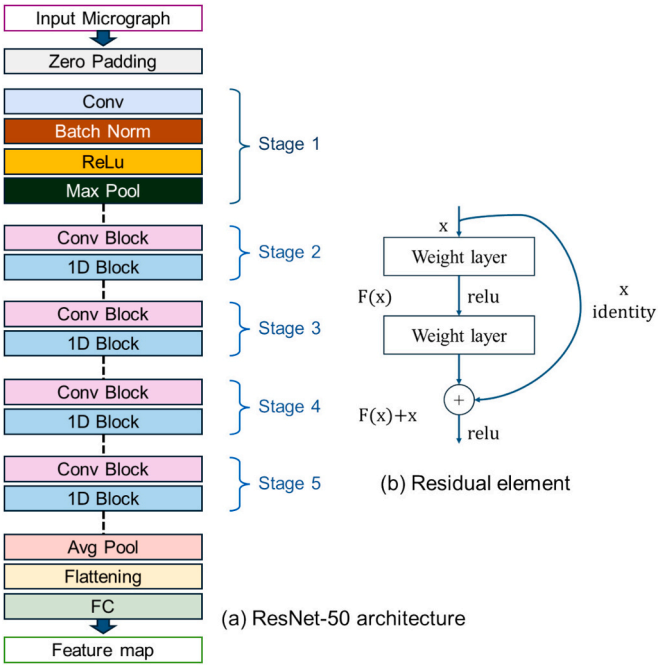


Fig. 4. (a) Architecture of ResNet50 containing an initial convolutional layer, five stage of residual blocks, and final layers for classification of phases. A Residual block starts with 1 × 1 convolution layer, followed by a 3 × 3 convolution layer and with 1 × 1 convolutional layer. The final output is then added to the input via a residual connection as shown in (b).

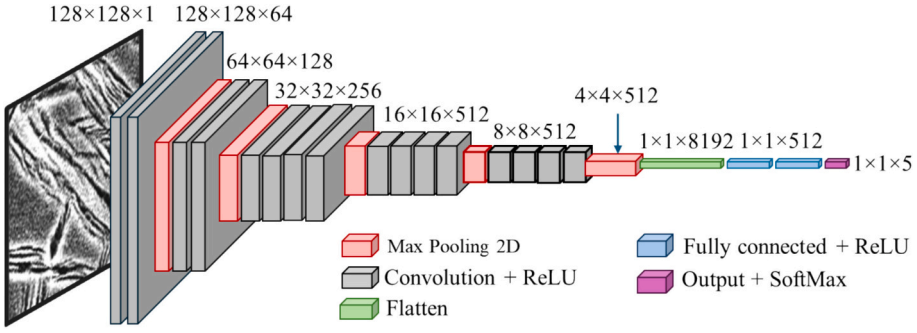


Fig. 3. The schematic of the VGG-19 model is used for the classification of patches belonging to five distinct phases, namely ferrite, pearlite, austenite, bainite, and martensite.

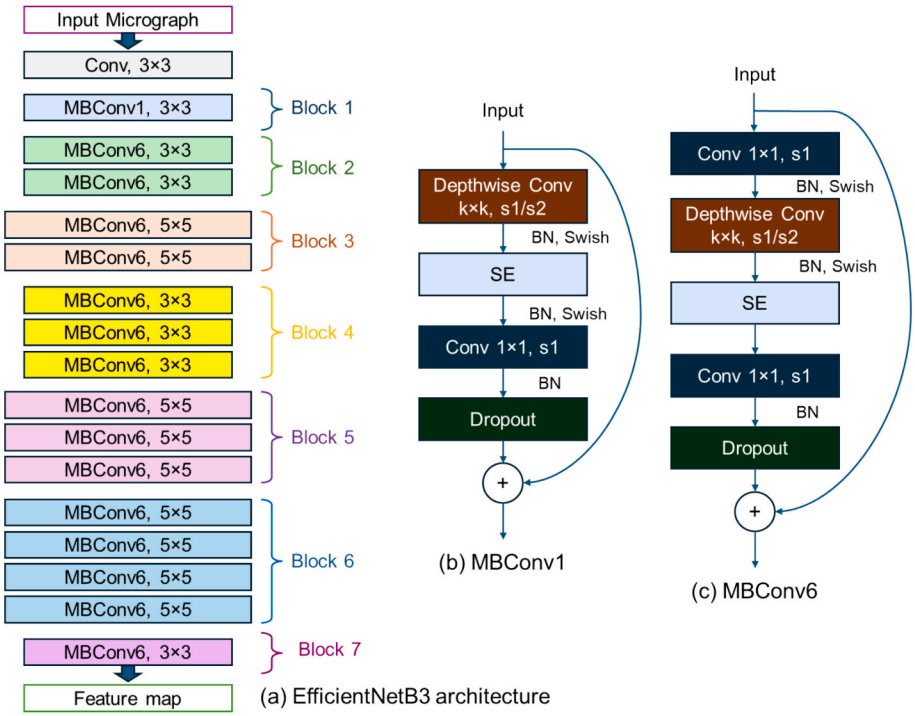


Fig. 5. (a) The schematic representation of EfficientNetB3 architecture consisting of MBConv1 and MBconv6 blocks used to extract feature maps. (b) A MBConv1 utilizes a 1 × 1convolution, depthwise convolution, squeeze and excitation module, and dropout layer, whereas MBConv6 adds a 3 × 3 depthwise convolution for increased non-linearity as shown in (c).

size to balance accuracy and efficiency. Compared to other architectures like VGG and ResNet variants, EfficientNet-B3 offers significantly higher accuracy per parameter and FLOP, making it well-suited for real-world applications, particularly in medical imaging (e.g., brain tumor and diabetic retinopathy classification) and mobile deployment where computational resources are limited. Its design demonstrates how carefully balanced scaling can yield superior performance without unnecessary model complexity.

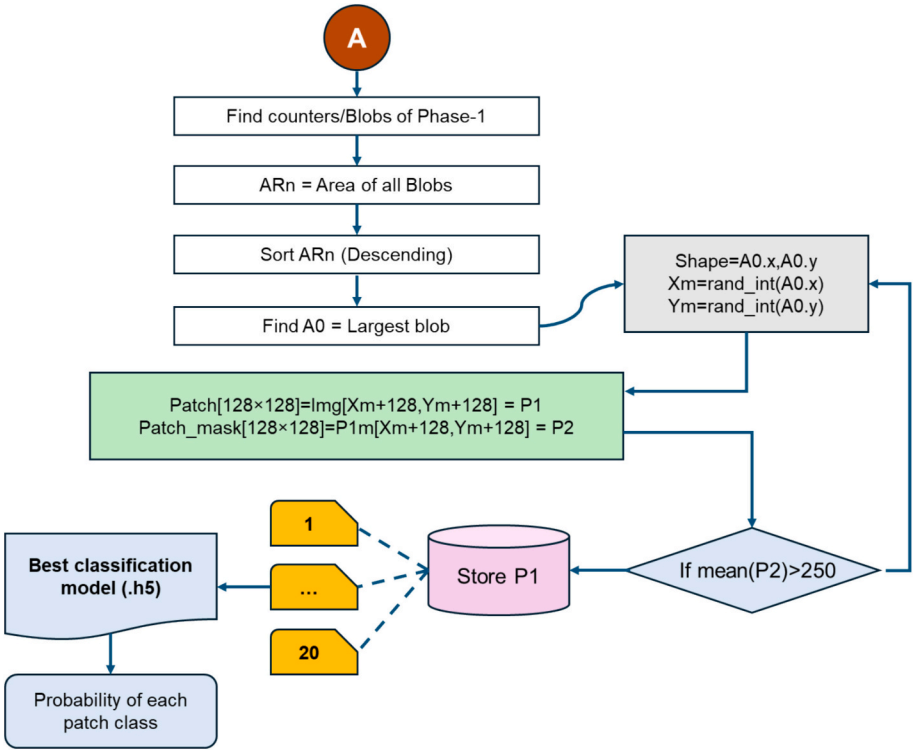


Fig. 6. The step-by-step procedure for extraction and classification of the patches. First the microstructure was segmented into distinct phases as depicted in Fig. 2. The contours/blobs were sorted in descending order of blobs' area. After that largest blob was used to extract the patches followed by the classification task was performed to classify the patches.

2.6. Computer vision of unseen micrographs

In the scope of evaluation, various steel microstructures were tested to validate the robustness of the proposed work. For the segmentation of an unseen microstructure, firstly Chan-Vese model was used to segment the image, after that extraction of the region proposals (i.e., foreground and background phases) was caried out (as discussed in section 2.3). Subsequently, area of all the blobs was calculated using morphological operation and sorted in descending order. Thereafter largest blob was extracted using largest bounding box function. In the next step, a series of patches were created from the largest blob of the segmentation. Each of these individual patches are independently classified using the best performing model and stored the probability obtained from the last layer of the classifier. The detailed representation of extraction of patches and prediction is shown in Fig. 6. The corresponding predictions of the patches from regions 1 and 2 then contribute accordingly to the final classification. The probability predictions are averaged to find the final phase segmentation results.

3. Results and discussion

3.1. Phase classification model development

Fig. 7(a & b) represent the accuracy and loss curves on the testing dataset of the phase classification models (i.e., VGG-19, Resnet-50, and EfficientnetB3) respectively. All these classification models are trained from scratch by setting up 'weights = None'. The accuracy obtained using VGG-19, Resnet-50, and EfficientnetB3 models was 85.8 %, 86.7 % and 82.2 % respectively. Also, the running time of each model, i.e., VGG-19, Resnet-50, and EfficientnetB3, was approximately 30m8s, 52m7s, and 70m18s respectively. Considering the accuracy and time for training the model, VGG-19 was selected as the classification model in our classification case of microconstituent patches. It was observed that the VGG19 model's accuracy does not improve much beyond 50 epochs, 0.4 dropout and 0.0001 learning rate. The VGG-19 model was trained using aforementioned settings which exhibited excellent training and testing performance, indicating that the model did not overfit the training and testing sets. The training of the classification model was performed on Intel(R) Xeon(R) Gold 5218R CPU, and Nvidia RTX A5000 GPU in Python-based TensorFlow 2.0 platform.

Fig. 8 exhibits the confusion matrix of the testing data for the classification obtained from the proposed VGG-19 model, which reveals the correct and wrong predictions for each of the microstructure phases of the steel. In general, the confusion matrix epitomizes the much deeper statistics about the type of error made by the classifier, if any. Alone, performance metrics like accuracy sometimes do not provide the complete information of the multiclass classification model's performance when the set of correct predictions in every class is not of equal size. For example, in this case, the number of correctly predicted instances for all five phases is uneven concerning their support in the training and testing datasets.

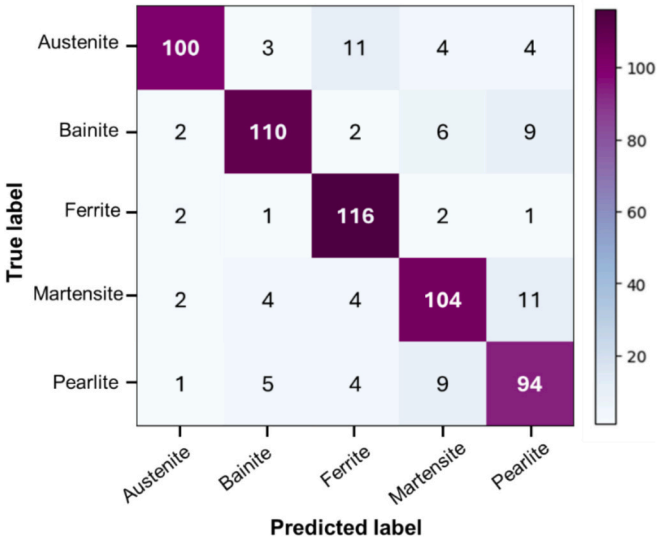


Fig. 8. The confusion matrix shows the overall performance based on the VGG-19 model on the testing data.

In the present context for the martensite phases, the predictability is very uncertain, unlike the other phases, namely, austenite, bainite, ferrite, and pearlite. Moreover, it can be found from Fig. 8 that the austenite, martensite, and pearlite phases are the most confused classes (f1-score is 82.6 %, 82.2 %, and 82.8 % respectively) to be predicted and may need repeated experimental confirmation for uncertainty, if any. Out of the 122 cases which are austenite in the testing dataset, the classification model correctly predicted 100 cases, and in the rest of 22 instances, 11 times the model was confused with ferrite phases. Same as in the case of the bainite phase, the model predicted the bainite 110 times correctly, 9 times it was predicted as pearlite, and 6 times it was predicted as martensite. Martensite was correctly predicted within the same class 104 times, but it was predicted as pearlite 11 times and sometimes as bainite and ferrite. In the case of pearlite, the model predicts it as pearlite most of the time, 9 times as martensite, and sometimes it is predicted to another class. Nonetheless, ferrite was the most accurately predicted class.

Fig. 9 (a) shows the instances where the VGG-19 model was confused austenite with ferrite in most cases. The misclassification of austenite as ferrite can often be attributed to overlapping microstructural characteristics, image resolution, or challenges in distinguishing between these phases. It may be because austenite and ferrite exhibit equiaxed grain

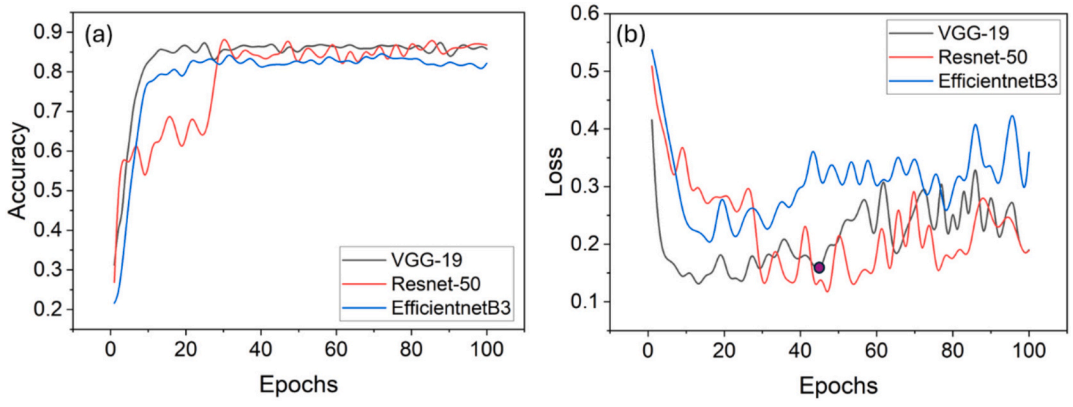


Fig. 7. The performance curve (a) accuracy and (b) loss for test data over 100 epoch iterations of the classification models (VGG-19, Resnet-50, and EffiecientnetB3).

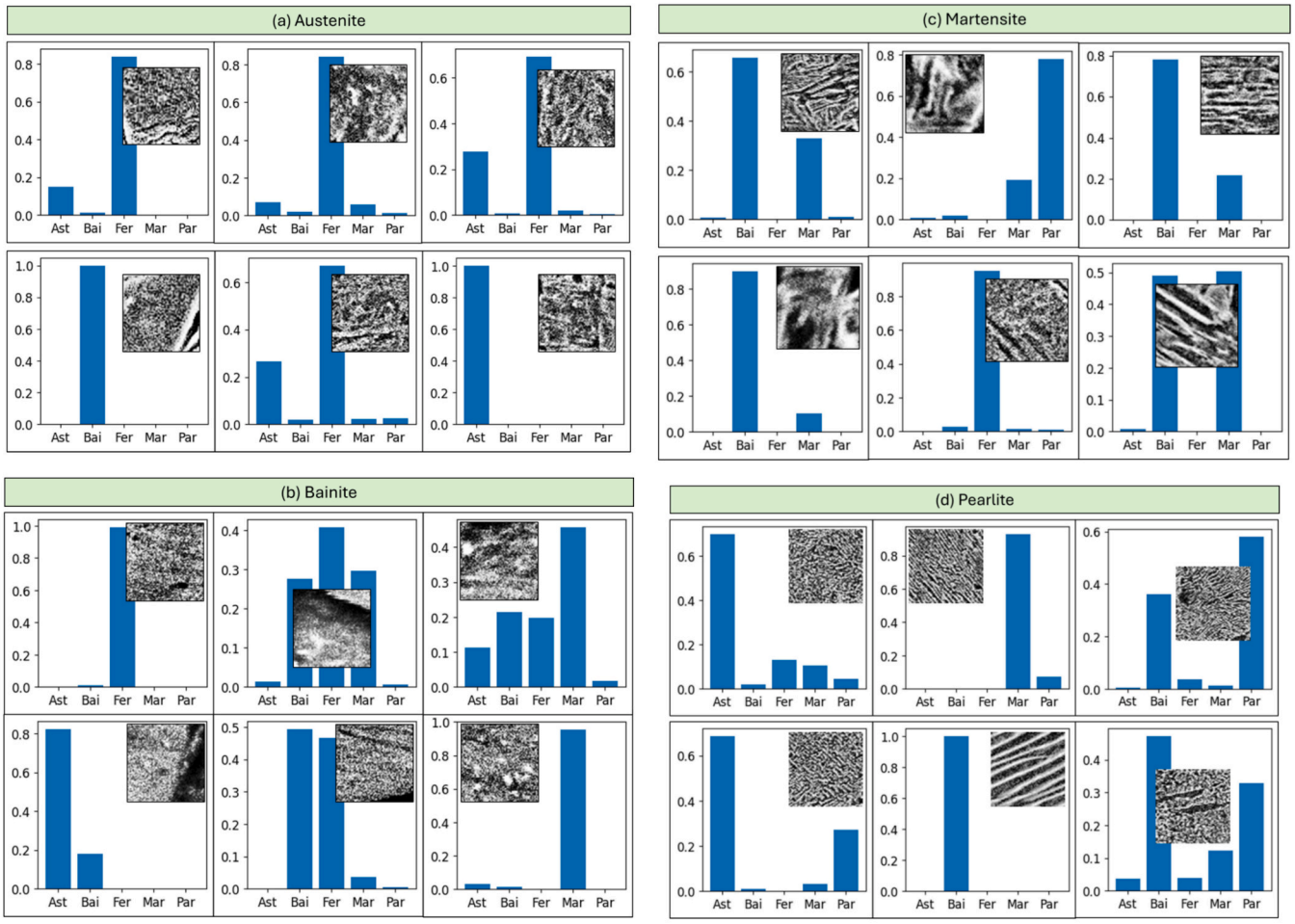


Fig. 9. The predicted probability density was obtained from the last layer of VGG-19 classification. In this fig. Major wrong prediction of the patches is depicted. (a) probability plot for wrong prediction of the austenite patches, (b) probability plot for wrong prediction of the bainite patches, (c) probability plot for wrong prediction of the martensite patches, and (d) probability plot for wrong prediction of the pearlite patches.

structures in micrographs. If austenite partially transforms into other phases during cooling (especially at intermediate cooling rates), residual austenite regions may coexist with martensite or bainite, leading to mixed features. This overlap can confuse the model, especially if retained austenite appears alongside martensite laths.

Fig. 9 (b) shows the example cases of wrong prediction of the bainitic phase patches. In most cases, bainite is confused with the pearlite phase, and subsequently with martensite. The misclassification of bainite as pearlite or martensite can be attributed to overlapping features in their microstructures and characteristics. The model predicted bainite as pearlite because both phases exhibit relatively low carbon content and appear similar in polished and etched micrographs at low-resolution images. Also, upper bainite can resemble pearlite due to its relatively coarser lath structure. Since bainite has a fine plate or acicular structure and martensite has a needle-like structure, without clear differentiation, bainitic ferrite may resemble lath martensite. The lower bainite can resemble martensite due to its finer structure and high dislocation density.

Fig. 9 (c) shows example cases of wrong prediction of the martensite phase patches. From a metallurgical perspective, the misclassification of martensite as bainite or pearlite can be explained by the overlapping microstructural features and transformations in steel. The needle-like appearance of martensite overlaps with the bainite's lath structure. Pearlite-like features are due to etching or tempering, can obscure martensite's sharp boundaries, making it appear lamellar.

Fig. 9 (d) shows the instances where the VGG-19 model was confused pearlite class with the other classes mainly as ferrite, bainite, and martensite. It may be because of the lamellar structure of pearlite, which can sometimes resemble martensite or bainite under specific imaging conditions due to the etching technique and imaging noise (visual overlap). Ferrite and pearlite may share similar grain structure under certain conditions, leading to confusion in feature extraction (Feature confusion). The patches generated from the higher resolution microstructure images might not provide sufficient details for accurate differentiation between martensite and bainite.

3.2. *Implementation of the classification and segmentation model*

For the prediction analysis of phase segmentation of the steel microstructure, a typical bainite-austenite microstructure has been considered as the first case study. The analysis reports comprising the ground truth of the microstructure (SEM and TEM images) along with intermediate and final predicted images are shown in Fig. 10. Fig. 10(a) shows the original SEM micrograph of the isothermally treated steel. The micrograph predominantly depicts the bainite structure with an untransformed austenitic region. The transmission electron microscopy (TEM) image of the same steel is shown in Fig. 10(a). The nanoscale bainite plates are clearly visible. This micrograph has been used in this investigation as a typical test case to validate the model with real in-house scale data. Fig. 10(b(i-viii)) represents the various stages of progression of image processing and segmentation results in the microstructure. It was found that both phases were accurately classified with

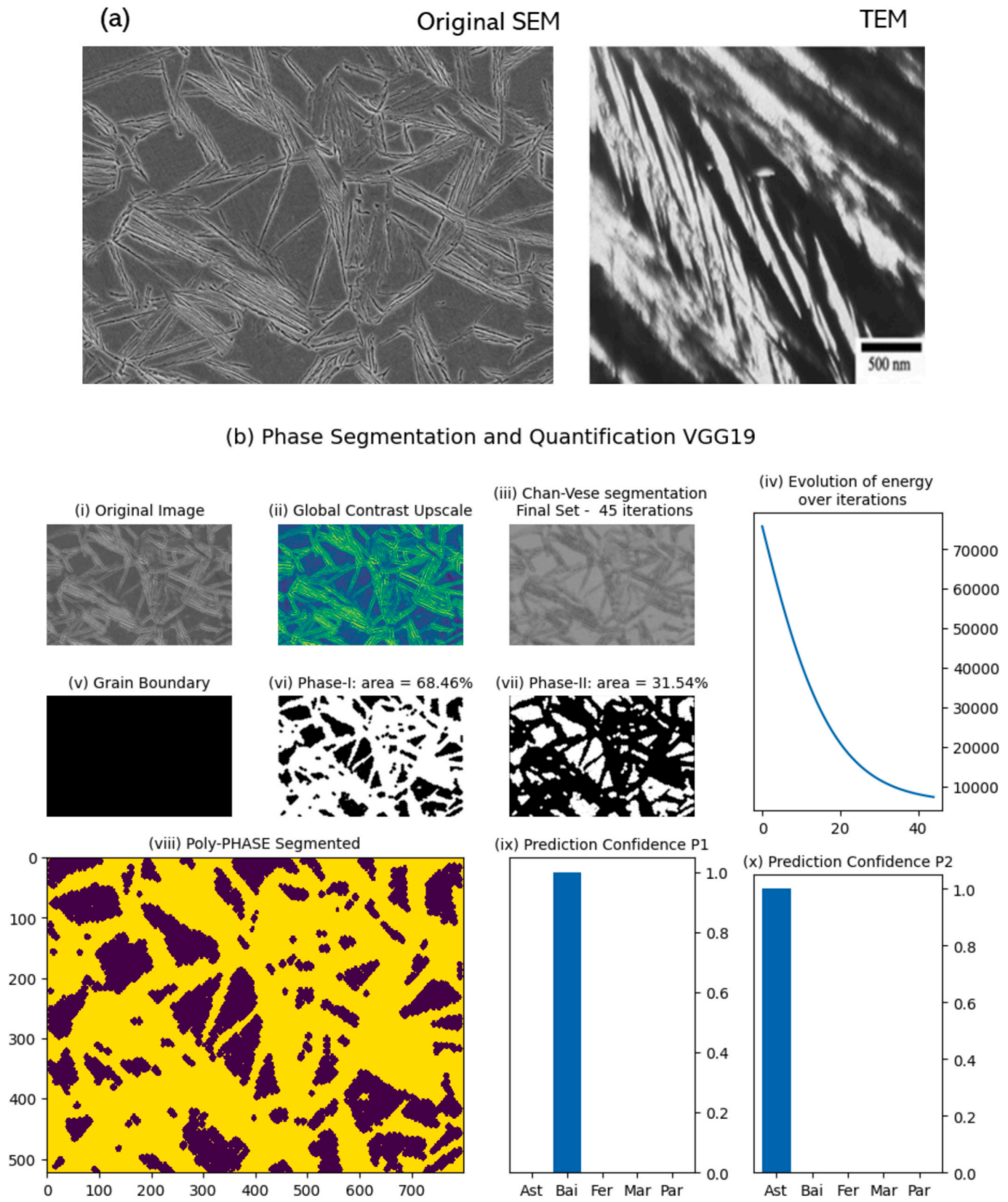


Fig. 10. Demonstration of prediction analysis of a typical case example with a predominantly bainitic-austenite type microstructure. The analysis report produce (a) SEM and TEM micrograph of the original bainite-austenite microstructure representing the ground truth, and (b) referring the sequence of segmented images: (i) original micrograph, (ii) global contrast upscaled, (iii) Chan-Vese (CV) final segmentation result, (iv) evolution of CV energy over 50 iterations, (v) grain boundary (vi and vii) phase masked. (viii) Final segmented microstructure, (ix and x) prediction probability of the identified phase.

the prediction probability of 100 % as shown in Fig. 10(b(ix & x)), which signifies high prediction accuracy. The quantitative examination of the polyphase segmentation resulted in covering area of bainite was 68.46 % and the covering area of the austenitic phase was 31.54 %.

Fig. 11(a) shows the original SEM micrograph of a sample of low-carbon steel, which contains 0.22 %wt. Carbon (C). The ferrite phase is primarily dominant over the pearlite phase in the micrograph. The TEM images of the structure, i.e., ferrite and pearlite of the above-mentioned steel, are shown in Fig. 11(a). The nanoscale of the ferrite is simply detectable, and the lamellar structure of the pearlite is clearly

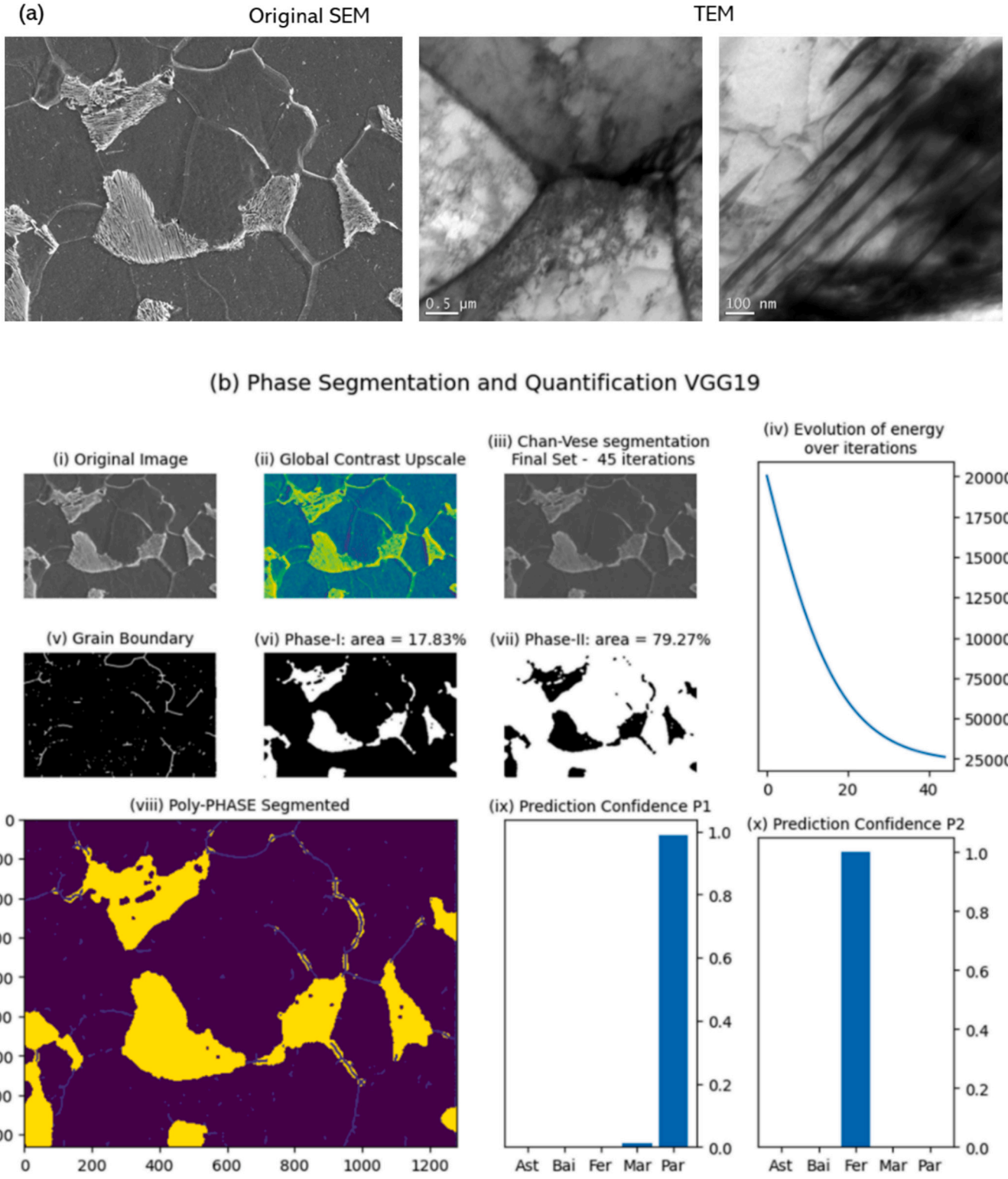


Fig. 11. Demonstration of prediction analysis of a typical case example with a predominantly ferrite-pearlite type microstructure. The analysis report produce (a) SEM and TEM micrograph of the original ferrite-pearlite microstructure representing the ground truth, and (b) referring the sequence of segmented images: (i) original micrograph, (ii) global contrast upscaled, (iii) Chan-Vese (CV) final segmentation result, (iv) evolution of CV energy over 50 iterations, (v) grain boundary (vi and vii) phase masked. (viii) Final segmented microstructure, (ix and x) prediction probability of the identified phase.

visible in the TEM image. This ferrite-pearlite-based SEM micrograph has been used as a second typical test case to validate the model with real in-house scale data.

Fig. 11(b (i)) shows the microstructure segmentation of the low carbon steel image, which was taken from the scanning electron microscopy at a magnification of 2000×. The experimental results obtained are grouped in Fig. 11(b) show that the pearlite and the ferrite phases are precisely segmented into their respective class. Fig. 11(b(ii-viii)) represents the various stages of progression of image processing and segmentation results in the microstructure. Ferrite is predicted

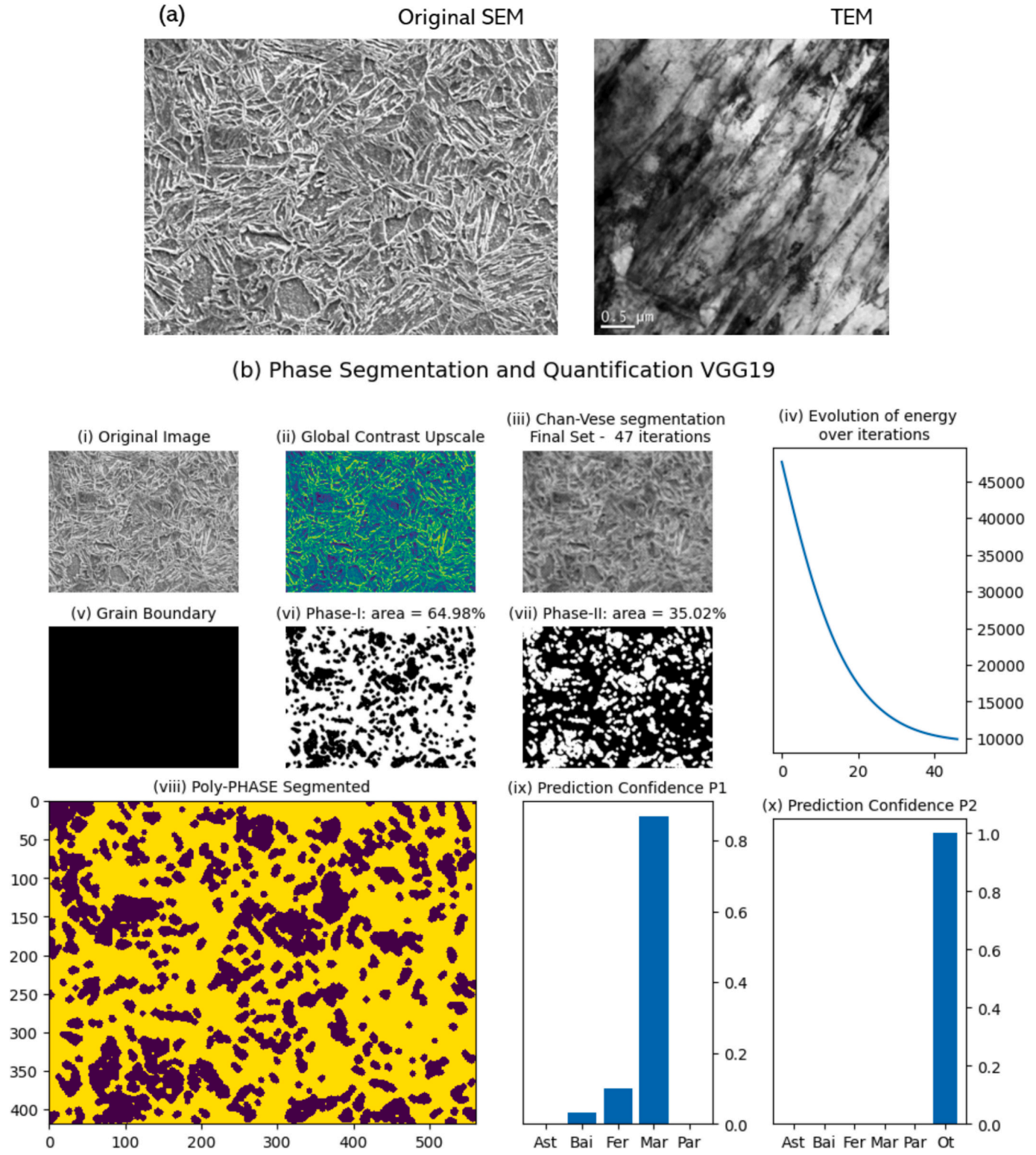


Fig. 12. Demonstration of prediction analysis of a typical case example with a predominantly martensite-bainite type microstructure. The analysis report produce (a) SEM and TEM micrograph of the original martensite-bainite microstructure representing the ground truth, and (b) referring the sequence of segmented images: (i) original micrograph, (ii) global contrast upscaled, (iii) Chan-Vese (CV) final segmentation result, (iv) evolution of CV energy over 50 iterations, (v) grain boundary (vi and vii) phase masked. (viii) Final segmented microstructure, (ix and x) prediction probability of the identified phase.

within the same phase, i.e., as ferrite, with a confidence of 100 %. However, in the case of pearlite, this phase was mostly predicted as pearlite and a little time as martensite. The probability of the pearlite phase prediction accuracy is 98 % as pearlite and 2 % as martensite, based on the average taken from the 10 patches captured from the different regions. Fig. 11(b(ix and x)), high probability of prediction of the phases using proposed model signifies appreciable accuracy in classification. In the quantitative analysis of poly-phase segmented image (Fig. 11b(viii)) found that the covering area of pearlite was 17.83 % and the covering area of the ferrite phase was 79.27 %. Some randomly distributed points in the pearlite (phase I) show the point holes/defects or polishing/etching defects on the surface.

Fig. 12(a) shows the in-house generated SEM micrograph of the continuously cooled low-carbon steel. The specimen was first austenitized at 950 °C, and then cooled at a constant cooling rate of 10 °C/s. It is noteworthy to mention here that the slow cooling rate causes retardation in the pearlite transformation and thereby allows the formation of bainite along with some amount of martensite (low-temperature transformation product). The nanoscale martensite is simply detectable as it shows a needle-like structure in the presented TEM image. This martensite-bainite-based SEM micrograph has been used as a third typical test case to validate the model with real in-house scale data.

For validation of the proposed model, a new combination of phases containing martensite-bainite was considered in this case. This micrograph was generated from a low-carbon steel, and the microstructure was captured at a magnification of 1000×. Fig. 12(b(i-viii)) represents the various stages of progression of image processing and segmentation results in the microstructure. The quantitative analysis of the polyphase segmented image resulted phase fractions of the martensite (phase I) and bainite (phase II) are 64.98 % and 35.02 % respectively. The segmentation results are grouped in Fig. 12(b), which shows that the martensite phase structures are precisely predicted, but bainite is not predictable. In Fig. 12(b(viii)), it can be found that the bainite phase is scattered like pieces throughout the micrograph. It is interesting to note here that that the bainite phase is separable in segmentation, but the largest bounding rectangle is not detected for phase II; thus, the patch size of 128 × 128 pixels (minimum size required for VGG-19 model as discussed in section 2.3) is not created for classification.

Fig. 13(a) illustrates the SEM micrograph of a hot-rolled specimen after inter-critical annealing at 750 °C followed by water quenching, showing martensite islands and ferrite matrix. The microstructure consists of a bright network of martensite islands dispersed in a dark ferritic matrix. The inter-critical annealing temperature of 750 °C, and the fast rate of cooling in a quenching medium of water are mainly responsible for the formation of martensitic islands in the ferrite matrix. The nanoscale needle-like structure of martensite plates is clearly visible in the TEM micrograph of the same steel, which is shown in Fig. 13(a). This micrograph is considered a typical case of dual-phase steel, which has been used to support and confirm the versatility of the model.

Fig. 13(b) shows the microstructure segmentation of the dual-phase steel image, which was taken from the scanning electron microscope at a magnification of 1000×. Fig. 13(b(i-viii)) represents the various stages of progression of image processing and segmentation results in the microstructure. The prediction probabilities of the phases are shown in Fig. 13(b(ix and x)), which depicted that the martensite and the ferrite are classified dominantly. The phase fractions of the martensite and ferrite phases are 40.35 % and 54.26 % respectively, and the remaining area belongs to the grain boundary. The prediction probability of the martensite and ferrite phase is approximately 64 % and 58 % respectively, which is higher than the probability of the other classes. The reason behind the low prediction accuracy of both phases is that patches created after the segmentation might consist of or overlap with other phases, which produces confusion for the model. It is interesting to note that the classification model is successfully learnt the information from the patches for recognizing the presence of ferrite and martensite phases with significantly high confidence level.

3.3. Robustness test and comparative analysis of micrographs

Fig. 14 exhibits the prediction obtained from the proposed VGG-19 model, which reveals the correct and wrong predictions for the various combinations of the phases in the microstructure of the two-phase steel. In the present context, ferrite-pearlite (FP), martensite-austenite (MA), bainite-austenite (BA), ferrite-ferrite (FF), martensite-ferrite (MF or DP) based two-phase steels have been taken to test the versatility of the model. In Fig. 14, “others” represents the two-phase system, which was either not possible or not used in this study (like martensite-pearlite, bainite-pearlite, pearlite-austenite, etc.) but predicted by the model. The model depicted 80.4 % accuracy on 235 types of different two-phase steel microstructures. It was found that MA-based microstructures were misclassified most of the time to different classes like FF, MF, and others. Moreover, it can be found from Fig. 14 that out of the 108 cases which are FP-based microstructure in the test data, the classification model correctly predicted 90 cases, and the rest of 18 cases, 2, 4, and 12 are confused with MA, FF, and other phases, respectively. Same as in the case of MF-based microstructure, the model predicted it as MF 39 times, 4 times it was predicted as MA, and 2 times it was predicted as FP microstructure.

Fig. 15 depicts the microstructure segmentation of the pearlite-ferrite based steel image, which was taken from the scanning electron microscopy at a magnification of 2000×. The results obtained are grouped in Fig. 15, which shows that ferrite phases are precisely predicted. Pearlite is misclassified as martensite with a confidence of approximately 52 % which is dominating than the prediction confidence of pearlite. It happens because the lamellar structure of pearlite resembles martensite under specific imaging conditions and imaging noise (visual overlap), leading to confusion in feature extraction (feature confusion). The patches generated from the higher resolution microstructure images might not provide sufficient details for accurate differentiation between pearlite and martensite. This problem may be encountered by creating more patches from different areas of the microstructure, but it increases computational time and resources. However, the phase fractions are accurately segmented by the Chan-Vese segmentation model, which provides the fraction area of phase I and phase II are 31.55 % and 62.06 % respectively.

3.4. Comparison of phase fraction based on the segmentation model

In this section, the segmentation results are verified and compared with the two popular open-source software, namely ImageJ and QuPath. For validation and comparison of the model, the pearlite fraction of the micrographs was compared with the above-mentioned two software ImageJ and QuPath. The results obtained from these methods and software are shown in Figs. 16 and 17. It can be observed from both Figs. 16 and 17 that our segmentation model generates balanced results between the QuPath and ImageJ phase segmentation.

Fig. 16 shows the segmentation results of the microstructure at the different magnifications of the ferrite and pearlite system image. In the case of segmentation results obtained using ImageJ, the clustering of the phases depends on the histogram of the brighter pixels. The brighter pixel of the image is levelled as a single phase, and the lower brighter image is considered as another phase. Thus, the pearlite phase fraction is lower in all cases as the magnification increases for ImageJ software. QuPath also segments the phases based on the pixel brightness, but it can cluster the same type of texture into a single phase. So, in QuPath software, the pearlite phase fraction covers more area. The percentage fraction of the pearlite is shown in Fig. 17. The pearlite phase fraction is higher at the lower magnification and consistent with slight variation over the other magnifications. The overall phase fraction of the pearlite is 20 ± 6, and the rest is the ferrite phase. The variation of the pearlite phase of ±6 is due to the focus and image captured during the SEM microscopy.

Fig. 18 shows the segmentation results of the microstructure at an

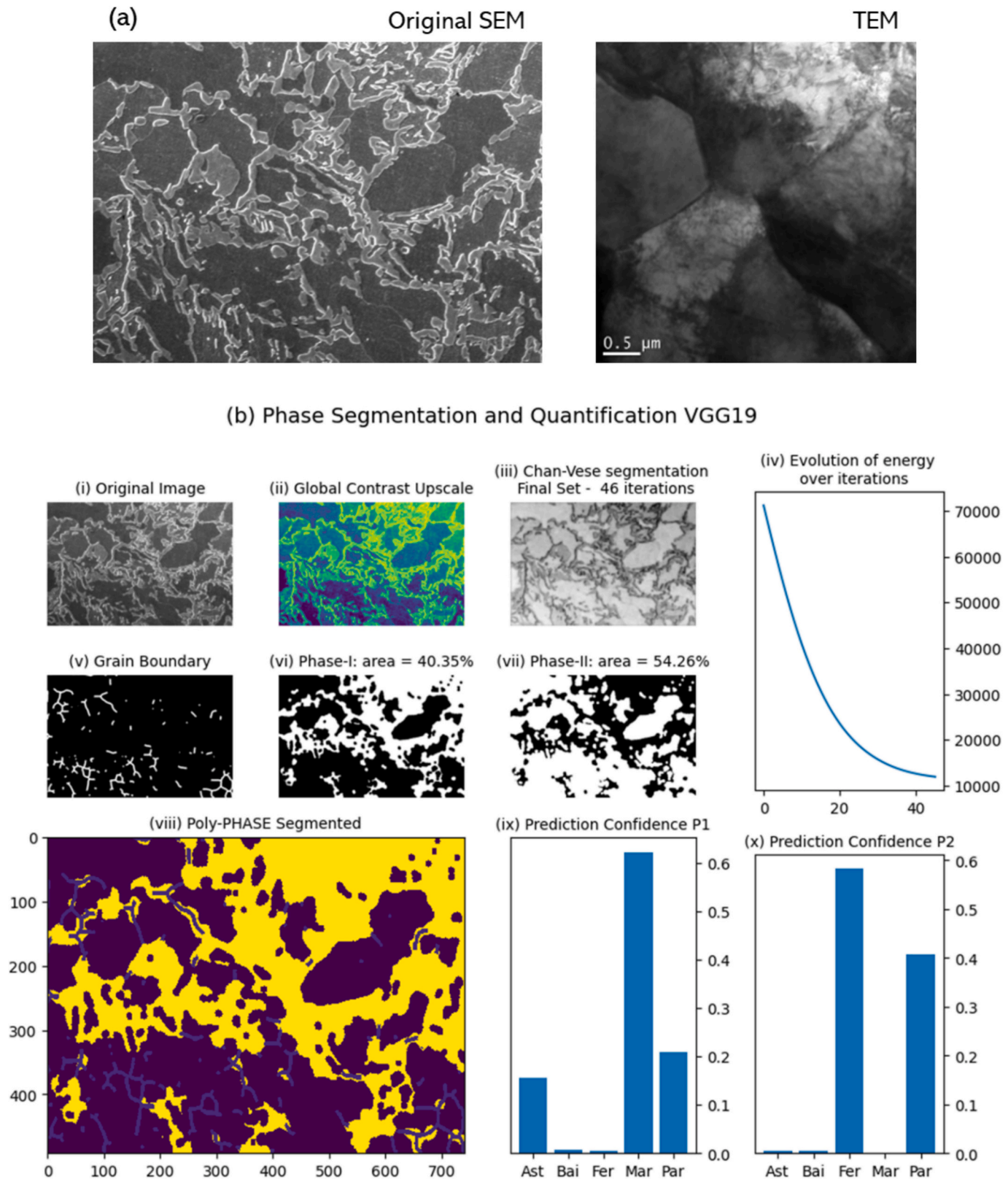


Fig. 13. Demonstration of prediction analysis of a typical case example with a predominantly ferrite-martensite type microstructure. The analysis report produce (a) SEM and TEM micrograph of the original ferrite-martensite microstructure representing the ground truth, and (b) referring the sequence of segmented images: (i) original micrograph, (ii) global contrast upscaled, (iii) Chan-Vese (CV) final segmentation result, (iv) evolution of CV energy over 50 iterations, (v) grain boundary (vi and vii) phase masked. (viii) Final segmented microstructure, (ix and x) prediction probability of the identified phase.

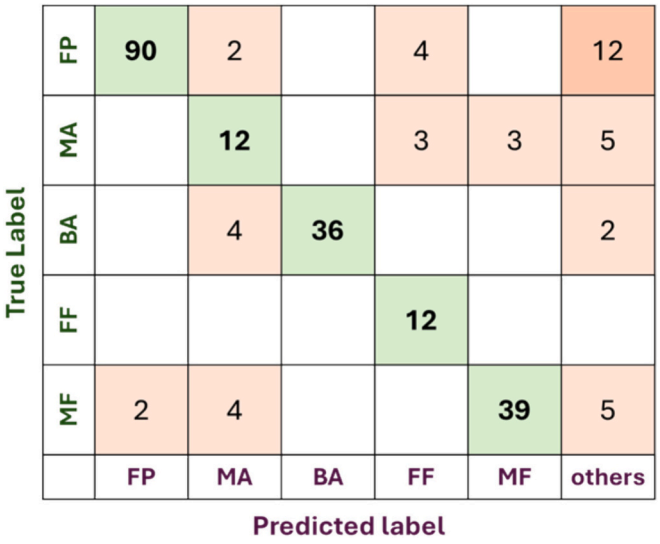


Fig. 14. The Confusion matrix represents the gross performance of true and wrong predictions of the proposed model on the various classes of microstructures.

increasing percentage of carbon at a fixed magnification of 2000× of the ferrite-pearlite system. It can be observed from Fig. 19 that the pearlite phase fraction is increased as the C (%wt.) increases, which is metallurgically valid. The pearlite fraction obtained from the increasing percentage of carbon in ImageJ is slightly low because ImageJ segments the phases based on the thresholding value. As we know, the pearlite phase is a combination of ferrite and cementite phases. Diffraction of light in ferrite and cementite can cause a pearl-like appearance due to the raised spacing of cementite acting as a diffraction grating. So, ImageJ only considered cementite and some areas of ferrite and grouped them into a single phase. So, as the pearlite phase increased, ImageJ was not able to detect the correct fraction of pearlite. In the case of the QuPath and Chan-Vese method, the pearlite phase is detected and clustered into a single phase very well, with a little deviation of ±2 % or ± 3 % approximately. The proposed automatic phase identification technique is significantly noteworthy because it quantifies phases with phase fraction variations within a realistic and acceptable range for the test materials over a broad range of steel SEM images.

Table 2 provides a comparative analysis of the ground truth annotations and the segmentation results generated by the Chan-Vese segmentation model. The evaluation was conducted using standard metrics, including pixel accuracy, Intersection over Union (IoU), and Dice similarity coefficient (DSC). The model achieved an overall pixel accuracy of 89.22 %, with mean IoU and Dice scores of 0.78 and 0.84, respectively,

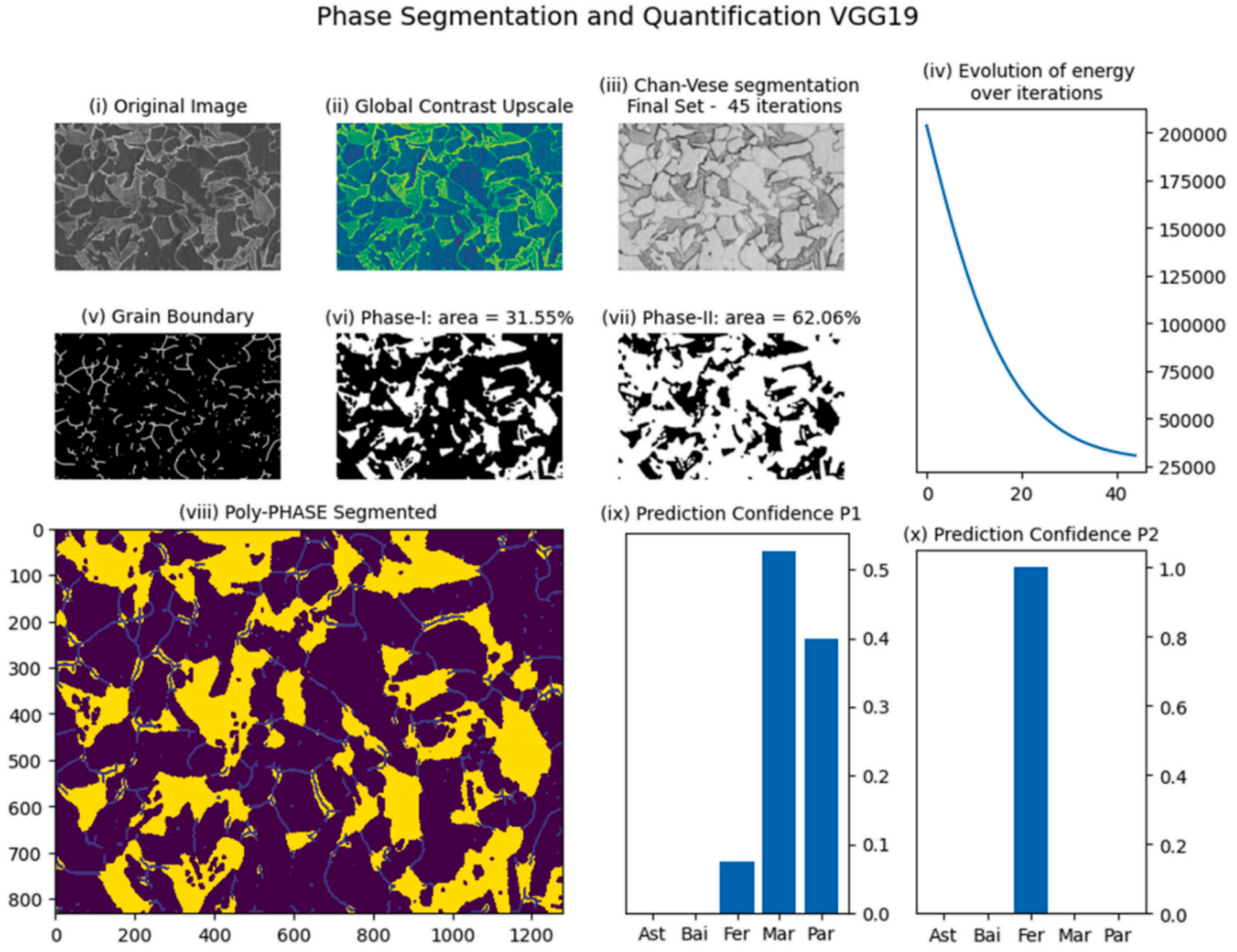


Fig. 15. A typical example of the heavily confused prediction made by the model, the micrograph belongs to a pearlite-ferrite based system but is predicted as a martensite-ferrite based system.

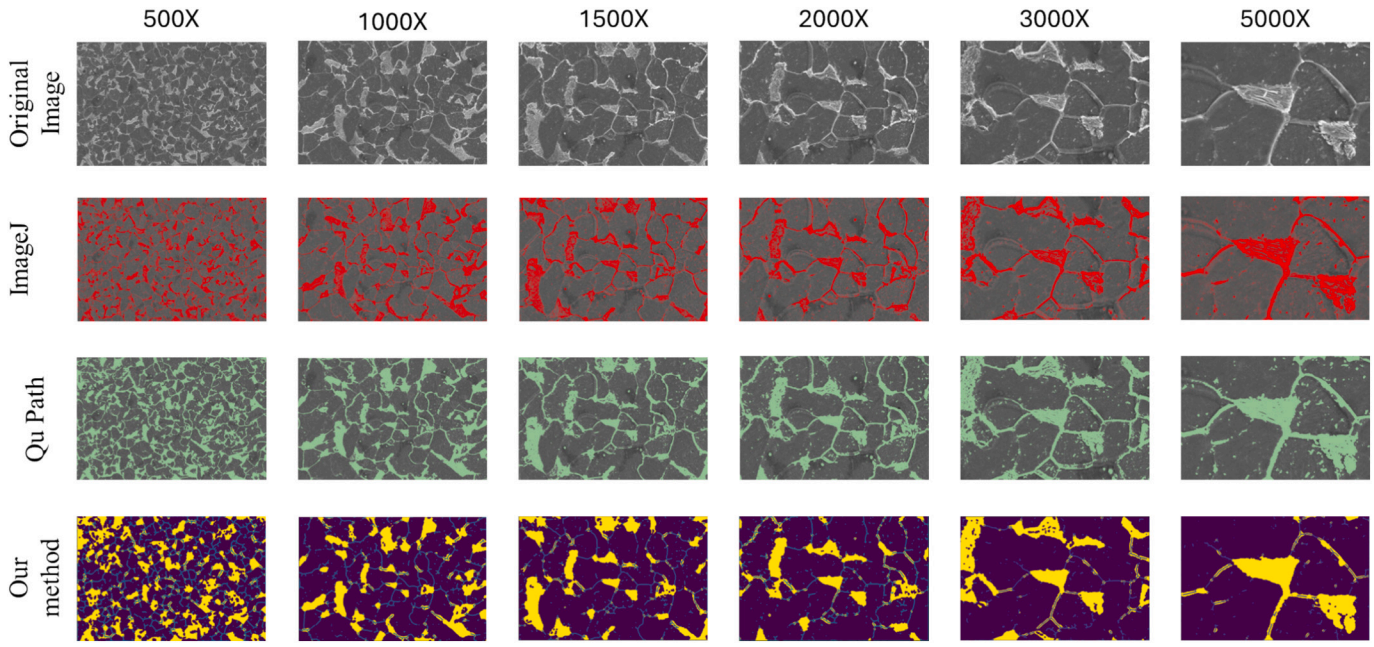


Fig. 16. Segmentation results are obtained from two popular open-source software namely ImageJ and QuPath and compared with the proposed model. From left to right magnification increases from 500× to 5000 × .

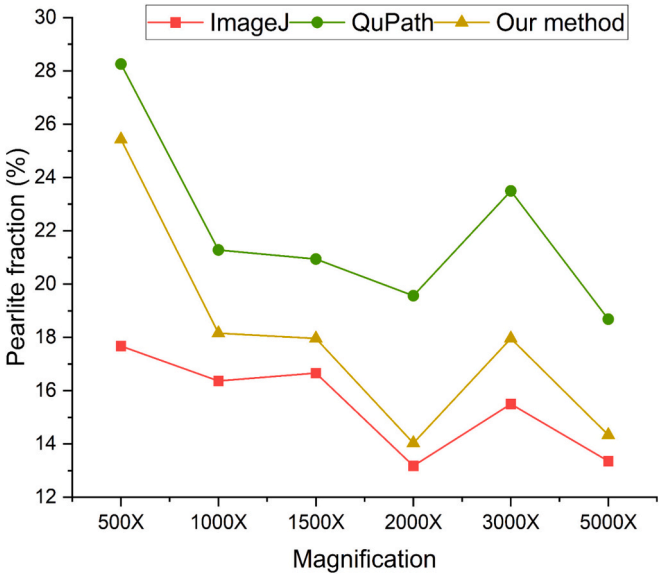


Fig. 17. Comparison of the pearlite phase fraction obtained from ImageJ, QuPath, and the proposed model with the increase of magnification.

across all classes. These values indicate a strong overlap between predicted regions and ground truth, confirming the reliability of the segmentation. Minor deviations were observed at object boundaries, which can be attributed to the complex morphological variations present in the dataset. Nevertheless, the quantitative results demonstrate that the proposed approach achieves a close approximation to expert annotations and can serve as a robust tool for automated microstructural characterization.

The matrices have been calculated on the microstructures given in Figs. 16 and 18.

3.5. Strategies to address phase confusion/misclassification and future directions

The confusion matrix (Fig. 8) revealed frequent misclassification among martensite, bainite, and pearlite, which is mainly consistent with their overlapping morphological features. While metallurgical explanations account for these similarities, several modeling strategies may further reduce such confusions. A hierarchical classification framework could first separate phases into broader categories (e.g., equiaxed versus lamellar/lath-like microstructures), followed by a specialized classifier to distinguish closely related phases such as bainite, martensite, and pearlite. Attention-based mechanisms (e.g., channel or spatial attention modules) could also be integrated into the backbone network to emphasize subtle but discriminative features, such as cementite lamellae in pearlite versus fine lath structures in bainite and martensite. In addition, the incorporation of phase-specific filters, such as oriented Gabor filters or morphological operators, may enhance the network's sensitivity to directionality and texture, thereby aiding in separating structurally similar phases. Finally, ensemble approaches that combine deep CNN features with handcrafted texture descriptors could provide complementary perspectives and reduce phase-specific biases. Although these approaches are beyond the scope of the current work, they present promising avenues for future studies aimed at improving phase-level discrimination.

4. Conclusions

We have demonstrated a framework on how to effectively use deep learning techniques and image segmentation algorithms in high-throughput analysis of steel microstructure, covering a variety of steel microstructural constituents. The study concludes the following

- The work created an efficient SEM micrograph database of advanced high-strength steels, which was effectively utilized to develop computer vision of phase constituents such as ferrite, pearlite, austenite, bainite, and martensite.
- The VGG-19, a CNN architecture-based trained model combined with Chan-Vese segmentation algorithms, effectively built the microstructure classification framework, enabling the digital vision

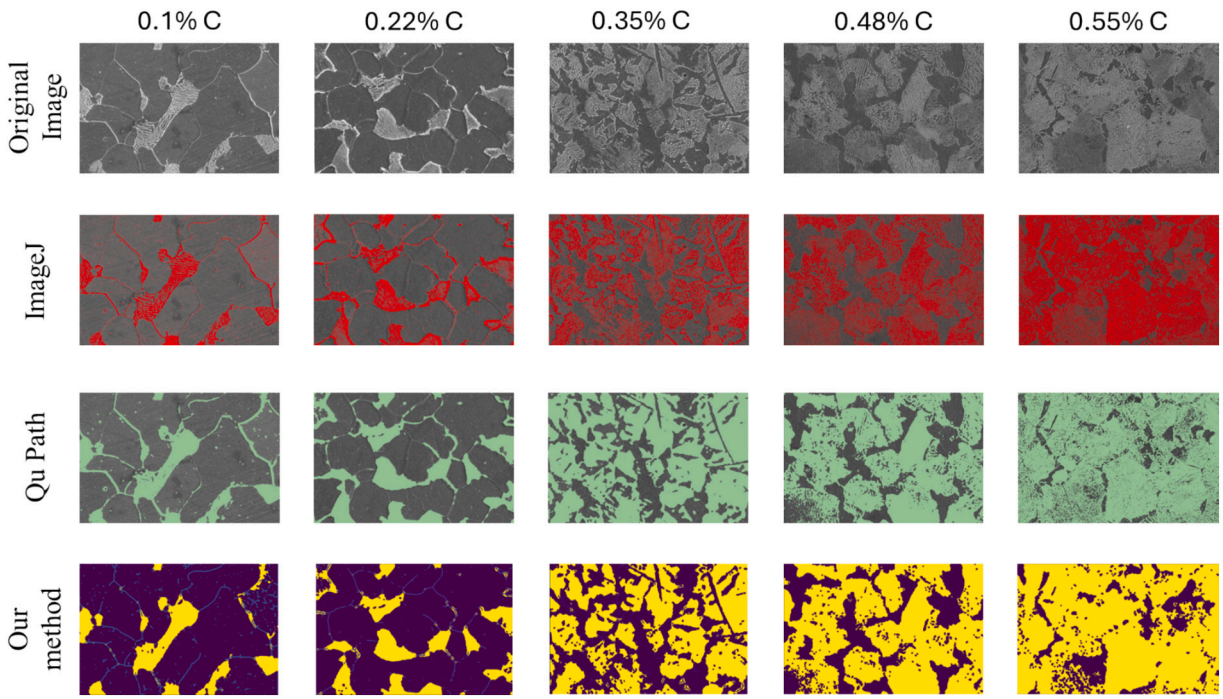


Fig. 18. Segmentation results are obtained from two popular open-source software, namely ImageJ and QuPath, and compared with the proposed model. From left to right, the carbon percentage increases from 0.1 % C to 0.55 % C.

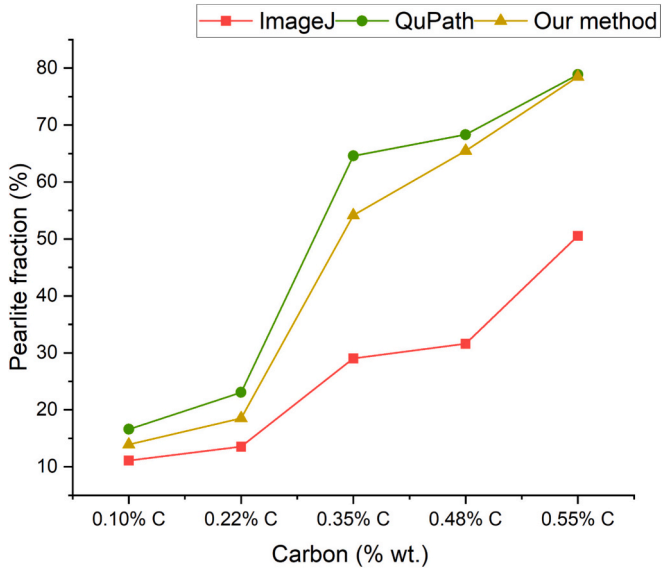


Fig. 19. Comparison of the pearlite phase fraction obtained from ImageJ, QuPath, and the proposed model with the increase of C (wt%).

Table 2
Quantitative comparison of ground truth and predicted segmentation of the ferrite-pearlite based micrographs, reported in terms of pixel accuracy, Intersection over Union (IoU), and Dice similarity coefficient (DSC).

Phase	IoU		F1-score		Accuracy	
	mean	S.D.	mean	S.D.	mean	S.D.
Ferrite	0.87	0.015	0.93	0.007	–	–
Pearlite	0.69	0.032	0.74	0.025	–	–
overall	0.784	0.018	0.84	0.014	89.22	1.67

of microstructure to show prediction accuracy on phase patches and complete microstructure annotation accuracy of 86 % and 80 % respectively. The extraction of phase patches using Chan-Vese segmentation as against the regular grid-based scheme allowed more selective patch extraction.

- Analysis of in-house generated benchmark data demonstrated that the phase volume fraction prediction accuracy is comparable with standard image analyzers such as ImageJ and QuPath. The sensitivity of phase prediction in a wide range of variation in resolution, variety of phase combinations, and changing phase fraction environment claimed generalization of proposed framework.
- The prediction result analysis depicted that the martensite is the most confusing phase, and it is frequently predicted as either bainite or pearlite, leading to the misclassification of the micrograph.
- Misclassification analysis results revealed that the grain size of the phases in the micrograph under study is too small with respect to the patch size (128 × 128 pixels) led to wrong leveling of the patch and thus misleading the training of the CNN model. This was noted as the primary limitation of the proposed model.

CRediT authorship contribution statement

Ashutosh Kumar Gupta: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Subir Gupta:** Visualization, Software, Data curation. **Swarup Kumar Ghosh:** Writing – review & editing, Project administration, Data curation. **Subhas Ganguly:** Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research is sponsored by ANRF, India, through its core research grant CRG/2021/005256.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.commatsci.2025.114319.

Data availability

The code and data are available on: https://github.com/akg0110/high-throughput-analysis-of-steel-microstructure/

References

[1] M. Rusanovsky, O. Beeri, G. Oren, An end-to-end computer vision methodology for quantitative metallography, Sci. Rep. 12 (2022), https://doi.org/10.1038/s41598-022-08651-w.
[2] Y. Zang, P.-S. Koutsourelakis, PSP-GEN: stochastic inversion of the process–structure–property chain in materials design through deep, generative probabilistic modeling, Acta Mater. 284 (2025) 120600, https://doi.org/10.1016/j.actamat.2024.120600.
[3] K. Li, B. DeCost, K. Choudhary, J. Hattrick-Simpers, A reproducibility study of atomistic line graph neural networks for materials property prediction, digital, Discovery 3 (2024) 1123–1129, https://doi.org/10.1039/d4dd00064a.
[4] J. Zhao, H. Yang, Y. Sui, Computer vision based automatic evaluation method of Y2O3 steel coating performance with SEM image, Sci. Rep. 15 (2025) 1722, https://doi.org/10.1038/s41598-024-85061-0.
[5] H. Ghauri, R. Tafreshi, B. Mansoor, Toward automated microstructure characterization of stainless steels through machine learning-based analysis of replication micrographs, J. Mater. Sci.: Mater. Eng. 19 (2024), https://doi.org/10.1186/s40712-024-00146-y.
[6] O. Haffner, E. Kučera, D. Rosinová, Applications of Machine Learning and Computer Vision in Industry 4.0, Applied Sciences (Switzerland) 14, 2024, https://doi.org/10.3390/app14062431.
[7] N. Kulesh, A. Bolyachkin, I. Suzuki, Y.K. Takahashi, H. Sepehri-Amin, K. Hono, Data-driven optimization of FePt heat-assisted magnetic recording media accelerated by deep learning TEM image segmentation, Acta Mater. 255 (2023), https://doi.org/10.1016/j.actamat.2023.119039.
[8] M. Göken, H. Vehoff, Quantitative metallography of structural materials with the atomic force microscope, Scr. Mater. 35 (1996) 983–989, https://doi.org/10.1016/1359-6462(96)00247-3.
[9] A.K. Gupta, S. Chakroborty, S.K. Ghosh, S. Ganguly, A machine learning model for multi-class classification of quenched and partitioned steel microstructure type by the k-nearest neighbor algorithm, Comput. Mater. Sci. 228 (2023) 112321, https://doi.org/10.1016/j.commatsci.2023.112321.
[10] C. Gorynski, M. Frei, F.E. Kruis, M. Winterer, Machine learning based quantitative characterization of microstructures, Acta Mater. 256 (2023) 119106, https://doi.org/10.1016/j.actamat.2023.119106.
[11] S. Lin, L. Xu, Z. Guo, D. Zhang, P. Zeng, Y. Tang, H. Pei, Superpixel-based principal feature clustering annotation method for dual-phase microstructure segmentation, Mater. Charact. 218 (2024), https://doi.org/10.1016/j.matchar.2024.114523.
[12] R. Tedoldi, J. Li, O. Engkvist, T. Pinaffo Almeida, S. Alipour Bonab, M. Yazdani-Asrami, A. Niyongabo Rubungo, K. Li, J. Hattrick-Simpers, A. Bousso Dieng, LLM4Mat-bench: benchmarking large language models for materials property prediction, Mach learn, Sci. Technol. 6 (2025) 020501, https://doi.org/10.1088/2632-2153/ADD3BB.
[13] B. Ma, J. He, A. Ramazani, N. Fehlemann, X. Wang, S. Münstermann, Irregular microstructure-property linkage for cast alloys by a novel deep learning approach: application on cast austenitic stainless steel, Mater Today Commun 35 (2023) 105979, https://doi.org/10.1016/j.mtcomm.2023.105979.
[14] H. Zhao, C. Wang, J. Fan, Predicting steel column stability with uncertain initial defects using bayesian deep learning, Appl. Soft Comput. 151 (2024) 111139, https://doi.org/10.1016/J.ASOC.2023.111139.
[15] Z. Xiong, Y. He, J.R. Hattrick-Simpers, J. Hu, Automated phase segmentation for large-scale X-ray diffraction data using a graph-based phase segmentation (GPhase) algorithm, ACS Comb. Sci. 19 (2017) 137–144, https://doi.org/10.1021/acscombsci.6b00121.
[16] S. Roskosz, J. Chrapoński, Ł. Madej, Application of systematic scanning and variance analysis method to evaluation of pores arrangement in sintered steel, Measurement 168 (2021) 108325, https://doi.org/10.1016/j.measurement.2020.108325.
[17] S. Banerjee, S.K. Ghosh, S. Datta, S.K. Saha, Segmentation of dual phase steel micrograph: an automated approach, Measurement 46 (2013) 2435–2440, https://doi.org/10.1016/j.measurement.2013.04.057.
[18] B.L. Decost, E.A. Holm, A computer vision approach for automated analysis and classification of microstructural image data, Comput. Mater. Sci. 110 (2015) 126–133, https://doi.org/10.1016/j.commatsci.2015.08.011.
[19] Z. Liu, L. Jin, J. Chen, Q. Fang, S. Ablameyko, Z. Yin, Y. Xu, A survey on applications of deep learning in microscopy image analysis, Comput. Biol. Med. 134 (2021) 104523, https://doi.org/10.1016/j.compbiomed.2021.104523.
[20] D.E. Rumelhart, G.E. Hinton, R.J. Williams, Learning representations by back-propagating errors, Nature 323 (1986) 533–536, https://doi.org/10.1038/323533a0.
[21] A.K. Shargh, C.D. Stiles, J.A. El-Awady, Deep learning accelerated phase prediction of refractory multi-principal element alloys, Acta Mater. 283 (2025) 120558, https://doi.org/10.1016/j.actamat.2024.120558.
[22] G.E. Hinton, S. Osindero, Y.W. Teh, A fast learning algorithm for deep belief nets, Neural Comput. 18 (2006) 1527–1554, https://doi.org/10.1162/NECO.2006.18.7.1527.
[23] S. Masubuchi, E. Watanabe, Y. Seo, S. Okazaki, T. Sasagawa, K. Watanabe, T. Taniguchi, T. Machida, Deep-learning-based image segmentation integrated with optical microscopy for automatically searching for two-dimensional materials, NPJ 2D Mater. Appl. 4 (2020) 3, https://doi.org/10.1038/s41699-020-0137-z.
[24] M. Ge, F. Su, Z. Zhao, D. Su, Deep learning analysis on microscopic imaging in materials science, Mater. Today Nano 11 (2020) 100087, https://doi.org/10.1016/J.MTNANO.2020.100087.
[25] E. Shelhamer, J. Long, T. Darrell, Fully convolutional networks for semantic segmentation, IEEE Trans. Pattern Anal. Mach. Intell. 39 (2014) 640–651, https://doi.org/10.1109/TPAMI.2016.2572683.
[26] D. García Peña, D. García Pérez, I. Díaz Blanco, J.M. Juárez, Exploring deep fully convolutional neural networks for surface defect detection in complex geometries, Int. J. Adv. Manuf. Technol. 134 (2024) 97–111, https://doi.org/10.1007/s00170-024-14069-7.
[27] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, A.L. Yuille, DeepLab: semantic image segmentation with deep convolutional nets, Atrous convolution, and fully connected CRFs, IEEE Trans. Pattern Anal. Mach. Intell. 40 (2018) 834–848, https://doi.org/10.1109/TPAMI.2017.2699184.
[28] V. Badrinarayanan, A. Kendall, R. Cipolla, SegNet: a deep convolutional encoder-decoder architecture for image segmentation, IEEE Trans. Pattern Anal. Mach. Intell. 39 (2017) 2481–2495, https://doi.org/10.1109/TPAMI.2016.2644615.
[29] H. Zhao, J. Shi, X. Qi, X. Wang, J. Jia, Pyramid scene parsing network, Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR (2017) 6230–6239, https://doi.org/10.1109/CVPR.2017.660.
[30] O. Ronneberger, P. Fischer, T. Brox, U-net: convolutional networks for biomedical image segmentation, Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 9351, 2015, pp. 234–241, https://doi.org/10.1007/978-3-319-24574-4_28.
[31] N. Chaurasia, S.K. Jha, S. Sangal, A novel training methodology for phase segmentation of steel microstructures using a deep learning algorithm, Materialia (Oxf) 30 (2023) 101803, https://doi.org/10.1016/J.MTLA.2023.101803.
[32] M.A.A. Hegazy, M.H. Cho, M.H. Cho, S.Y. Lee, U-net based metal segmentation on projection domain for metal artifact reduction in dental CT, Biomed. Eng. Lett. 9 (2019) 375–385, https://doi.org/10.1007/s13534-019-00110-2.
[33] Z. Zhang, J. Hu, R. Wang, X. Chen, D. Yang, V.P. Vavilov, Y. Duan, H. Zhang, Automatic segmentation of microporous defects in composite film materials based on the improved attention U-net module, Quant. InfraRed Thermogr. J. (2024) 1–16, https://doi.org/10.1080/17686733.2024.2387406.
[34] A. Haldar, S. Suwas, D. Bhattacharjee, eds., Microstructure and Texture in Steels, Springer, London, London, 2009, https://doi.org/10.1007/978-1-84882-454-6.
[35] T.B. Hilditch, T. de Souza, P.D. Hodgson, Properties and automotive applications of advanced high-strength steels (AHSS), in: Welding and Joining of Advanced High Strength Steels (AHSS), Elsevier Inc., 2015, pp. 9–28, https://doi.org/10.1016/B978-0-85709-436-0.00002-3.
[36] R. Song, D. Ponge, D. Raabe, R. Kaspar, Microstructure and crystallographic texture of an ultrafine grained C-Mn steel and their evolution during warm deformation and annealing, Acta Mater. 53 (2005) 845–858, https://doi.org/10.1016/j.actamat.2004.10.051.
[37] H. Hu, T. Zhao, Z. Ning, J.-F. Wen, T. Shen, S. Wang, M. Song, A novel age-hardenable austenitic stainless steel with superb printability, Acta Mater. 283 (2025) 120547, https://doi.org/10.1016/j.actamat.2024.120547.
[38] B.L. DeCost, B. Lei, T. Francis, E.A. Holm, High throughput quantitative metallography for complex microstructures using deep learning: a case study in ultrahigh carbon steel, Microsc. Microanal. 25 (2019) 21–29, https://doi.org/10.1017/S1431927618015635.
[39] A.R. Durmaz, M. Müller, B. Lei, A. Thomas, D. Britz, E.A. Holm, C. Eberl, F. Mücklich, P. Gumbsch, A deep learning approach for complex microstructure inference, Nat. Commun. 12 (2021), https://doi.org/10.1038/s41467-021-26565-5.
[40] X. Liu, C. Aldrich, Automatic Microstructural Classification of Ultrahigh Carbon Steel with Vision Transformers and Convolutional Neural Networks, in: IFAC-PapersOnLine, Elsevier B.V, 2024, pp. 119–123, https://doi.org/10.1016/j.ifacol.2024.09.301.
[41] K. Tsutsui, H. Terasaki, K. Uto, T. Maemura, S. Hiramatsu, K. Hayashi, K. Moriguchi, S. Morito, A methodology of steel microstructure recognition using SEM images by machine learning based on textural analysis, Mater Today Commun 25 (2020) 101514, https://doi.org/10.1016/J.MTCOMM.2020.101514.
[42] S. Gupta, A. Banerjee, J. Sarkar, M. Kundu, S.K. Sinha, N.R. Bandyopadhyay, S. Ganguly, Modelling the steel microstructure knowledge for in-silico recognition of phases using machine learning, Mater. Chem. Phys. 252 (2020) 123286, https://doi.org/10.1016/j.matchemphys.2020.123286.

[43] S.M. Azimi, D. Britz, M. Engstler, M. Fritz, F. Mücklich, Advanced steel microstructural classification by deep learning methods, Sci. Rep. 8 (2018) 1–14, https://doi.org/10.1038/s41598-018-20037-5.
[44] B.I. Bachmann, M. Müller, D. Britz, T. Staudt, F. Mücklich, Reproducible quantification of the microstructure of complex quenched and quenched and tempered steels using modern methods of machine learning, Metals (Basel) 13 (2023), https://doi.org/10.3390/met13081395.
[45] M. Müller, D. Britz, L. Ulrich, T. Staudt, F. Mücklich, Classification of bainitic structures using textural parameters and machine learning techniques, Metals (Basel) 10 (2020), https://doi.org/10.3390/met10050630.
[46] M. Müller, D. Britz, T. Staudt, F. Mücklich, Microstructural classification of bainitic subclasses in low-carbon multi-phase steels using machine learning techniques, Metals (Basel) 11 (2021), https://doi.org/10.3390/met11111836.
[47] B.L. DeCost, T. Francis, E.A. Holm, Exploring the microstructure manifold: image texture representations applied to ultrahigh carbon steel microstructures, Acta Mater. 133 (2017) 30–40, https://doi.org/10.1016/j.actamat.2017.05.014.
[48] S. Saifullah, R. Drežewski, Modified histogram equalization for improved CNN medical image segmentation, Procedia Comput. Sci 225 (2023) 3021–3030, https://doi.org/10.1016/j.procs.2023.10.295.
[49] T.F. Chan, L.A. Vese, Active contours without edges, IEEE Trans. Image Process. 10 (2001) 266–277, https://doi.org/10.1109/83.902291.
[50] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778, https://doi.org/10.1109/CVPR.2016.90.
[51] M. Tan, Q.V. Le, EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks, 2019, pp. 6105–6114, in: https://proceedings.mlr.press/v97/tan19a.html. accessed August 31, 2025.